AMPZ

Neural Amp Modeler for iOS

User Manual

Version 1.9.0 — October 2026

LNH Enterprises LLC

Contents

1 Getting Started 7
- 1.1 What is Ampz? 7
- 1.2 System Requirements 7
- 1.3 First Launch & Audio Permissions 8
- 1.4 Connecting Your Guitar 8
- 1.5 Quick Start — Playing in 60 Seconds 8

2 The Main Screen 10
- 2.1 Row 1: Top Toolbar 10
- 2.2 Row 2: Effects Chain Strip 12
- 2.3 Row 3: Algorithm Bar 13
- 2.4 Row 4: Parameter Area 14

3 Signal Path: Split Lanes & the Mix Block 15
- 3.1 Overview 15
- 3.2 Chain Overview 15
- 3.3 Signal Path Templates 16
- 3.4 Splitting by Hand 18
- 3.5 The Mix Block 18
- 3.6 Lane Tags 19

4 NAM Amplifier 20
- 4.1 What is Neural Amp Modeling? 20
- 4.2 Getting NAM Models 20
- 4.3 Model Browser 20
- 4.4 NAM Controls 27

5 Effects Reference 29
- 5.1 Dynamics 29
- 5.2 Distortion / Overdrive 30
- 5.3 Modulation 33
- 5.4 Pitch 36
- 5.5 Wah 37
- 5.6 EQ / Filters 39
- 5.7 Reverb 41
- 5.8 Delay 43
- 5.9 IR Loader 44

6 Preset System 48
- 6.1 100-Slot Architecture 48
- 6.2 Preset Browser 48
- 6.3 Saving Presets 49
- 6.4 What Gets Saved 50

6.5 Sharing Rigs 50
7 Scenes 52
7.1 Overview 52
7.2 What a Scene Stores 52
7.3 Switching and Editing 52
7.4 Renaming 52
8 Looper 53
8.1 Overview 53
8.2 Dual-Loop Architecture 53
8.3 Recording Flow 53
8.4 Loop Modes 54
8.5 Routing Modes 54
8.6 Loop Controls 54
8.7 Volume Control 54
9 Drum Loops 55
9.1 Overview 55
9.2 Loop Browser 55
9.3 Genres & Loop Count 56
9.4 Transport Controls 56
9.5 Integration 56
10 Tuner 57
10.1 Overview 57
10.2 Display 57
10.3 Controls 57
11 Live Performance Mode 59
11.1 Overview 59
11.2 Layout 59
11.3 Effect Abbreviations 61
11.4 Screen Lock Prevention 61
12 MIDI 62
12.1 Overview 62
12.2 Enabling MIDI 62
12.3 Device Discovery 63
12.4 Bluetooth MIDI 63
12.5 Active Preset 63
12.6 Program Change 65
12.7 CC Mappings 65
12.8 Slot Bypass Mappings 65
12.9 Stomp Mappings 67
12.10 Global Effect Mappings 68
12.11 Action Mappings 70
12.12 MIDI Learn 72
13 Settings 73
13.1 Audio Configuration 73

13.2 Background Audio 74
13.3 Effect Trails (Spillover) 74
13.4 MIDI Settings 75
13.5 Open Source Licenses 75
14 Tips & Troubleshooting 76
14.1 Signal Chain Best Practices 76
14.2 Latency Optimization 76
14.3 Common Issues 77
15 The Ampz AUv3 Plugin 78
15.1 What the Plugin Includes 78
15.2 Loading Ampz in a Host 78
15.3 Sharing Models, IRs, and Presets 78
15.4 Host Automation 79
15.5 Saving Plugin State 79

Quick Reference Card

The Ampz main screen is organized into four horizontal rows. Here is a map of every control:

Figure 1: Ampz Main View

Top Toolbar (Row 1) — Navigation and utilities

#	Control	Function
1	AMPZ	App logo / home indicator
2	< >	Navigate presets (previous / next)
3	Preset number + name	Tap to open Preset Browser
4		Save current settings to a preset slot
5	120 BPM	Shows current tempo; tap green LED to set BPM
6		Toggle Drum Machine overlay
7		Toggle Looper overlay
8		Toggle Live Performance mode
9		Toggle Settings overlay (MIDI Settings lives inside)

#	Control	Function
10		Toggle Tuner overlay
11		Global bypass — disables all processing

Meters and Scenes — Under the toolbar: input and output meters with the **DSP** load badge (tap it for the per-pedal DSP Performance panel), and the **scene row** of eight capsules (tap to switch, long-press to rename). On short phones the scene row folds into an “S” menu in the toolbar.

Effects Chain Strip (Row 2) — 16 drag-to-reorder slots between the **IN** and **OUT** tabs

Each slot is color-coded by effect category. Tap to select, double-tap to open the Effect Browser, flick up or down to bypass, drag to reorder. Tap **IN** for the Chain Overview and the Signal Path templates. A split rig shows its two lanes as stacked rows with the Split node and the grey **Mix** pedal.

Algorithm Bar (Row 3) — Shows the selected algorithm for the active effect

Parameter Area (Row 4) — Knobs, sliders, and toggles for the active effect

1 Getting Started

1.1 What is Ampz?

Ampz is a professional-grade guitar amp modeling app for iOS, powered by Neural Amp Modeling (NAM) technology. NAM uses deep neural networks (WaveNet and LSTM architectures) trained on real amplifier recordings to faithfully reproduce the sonic characteristics of physical guitar amplifiers — at a fraction of the cost and weight.

Ampz is designed for ultra-low latency performance. Using Apple's RemoteIO Audio Unit with 64-sample buffers at 48 kHz, Ampz achieves buffer latency as low as 1.33 ms — making it feel as responsive as a real amplifier.

Key Features:

- Real-time neural amp modeling with WaveNet/LSTM inference
- 16-slot effects chain with 17 effect types, any slot can be a NAM amp
- Split signal paths: two parallel lanes with a Mix block, Signal Path templates, dual-amp and dual mono rigs
- 8 scenes per preset for gapless sound changes inside one rig
- 100-slot preset system (factory + user presets)
- Dual-loop looper with circular waveform display
- Built-in drum machine with 6 pattern categories
- Chromatic tuner with adjustable reference pitch
- MIDI control with CC mapping and MIDI Learn
- Live performance mode with scene pads and ten assignable stomps
- Tone3000 integration for downloading community amp models

1.2 System Requirements

Requirement	Details
iOS Version	18.0 or later
Device	iPhone (landscape) or iPad — iPad uses an adaptive sidebar layout with scaled controls
Audio Interface	USB or Lightning compatible audio interface
Storage	50 MB for app + NAM models

TIP: A USB or Lightning audio interface with an instrument-level input is required for the best latency and signal quality.

1.3 First Launch & Audio Permissions

When you first open Ampz, iOS will request audio input access. **You must grant this permission** — Ampz needs audio input to process your guitar signal.

1. Tap **“Allow”** when the audio input permission dialog appears
2. Ampz will automatically configure the audio session for low-latency processing
3. The green LED in the toolbar pulses to confirm audio is running

1.4 Connecting Your Guitar

1. Connect your audio interface to your iPhone via USB or Lightning adapter
2. Plug your guitar into the interface’s instrument input
3. Open Settings (gear icon) to select the correct input channel
4. Adjust the input gain on your interface — aim for a strong signal without clipping

NOTE: Ampz supports multi-channel USB and Thunderbolt audio interfaces. Use the Settings overlay to select which input and output channels to use.

1.5 Quick Start — Playing in 60 Seconds

1. Connect your guitar via a USB audio interface
2. Open Ampz — the NAM Amplifier slot is ready in the effects chain
3. Download a NAM model: tap the NAM slot, switch to the **Tone3000** tab, sign in, and download any amp model — or import a `.nam` file from your device via the **My Models** tab
4. Play a few notes — you should hear the amp model through your guitar signal
5. Add effects by double-tapping empty slots in the effects chain
6. Use the preset chevrons (‹ ›) to browse presets, or tap the preset name to open the Preset Browser
7. Tap the bypass button (⏻) to toggle global bypass

Figure 2: The Ampz main screen with all controls visible

2 The Main Screen

The Ampz interface is a single-screen design organized into four horizontal rows, optimized for quick access during performance.

Figure 3: Main screen — four horizontal rows

2.1 Row 1: Top Toolbar

The top toolbar contains all navigation and utility controls in a compact 58-pixel strip.

2.1.1 Logo & Preset Controls

- **Logo** (AMPZ) — App identity marker, always visible at the left edge.
- **Preset Navigation** — Chevron left (‹) and chevron right (›) step through the 100 preset slots sequentially.
- **Preset Display** — Shows the current preset number (two digits, e.g., “00”) and name. Tap the display area to open the full Preset Browser.
- **Save** (⏻) — Opens the Save Preset dialog to store the current effects chain state to any user slot.

2.1.2 BPM & Tap Tempo

- **BPM Display** — Shows the current global tempo as a number (e.g., “120 BPM”).
- **Tap Tempo LED** — A pulsing green circle that flashes in time with the current BPM. Tap it repeatedly to set the tempo by tapping.

The global tempo drives tempo-synced effects (delay, tremolo, chorus), the drum machine, and the looper’s quantize feature.

2.1.3 Utility Buttons

Icon	Name	Function
	Drum Machine	Opens/closes the Drum Machine overlay
	Looper	Opens/closes the Looper overlay
	Live Mode	Enters/exits Live Performance mode
	Settings	Opens/closes the Settings overlay
	Tuner	Opens/closes the Tuner overlay
	Bypass	Toggles global bypass (all processing off/on)

When a utility button is active (its overlay is open), the icon highlights in the app's accent color (red). Only one overlay can be open at a time — tapping a different utility button closes the current overlay and opens the new one. MIDI Settings ( ) is reached from inside the Settings overlay.

2.1.4 Meter Strip & DSP Badge

Below the toolbar, the meter strip shows the input and output levels with a **DSP** badge that reads the chain's processing load as a percentage of the audio buffer. Tap the badge to open the **DSP Performance** panel: one row per pedal with its CPU cost, plus a benchmark that times every NAM model in your library one at a time and restores your rig afterwards. Use it to see which pedal, or which amp capture, is the expensive one before a rig starts to crackle.

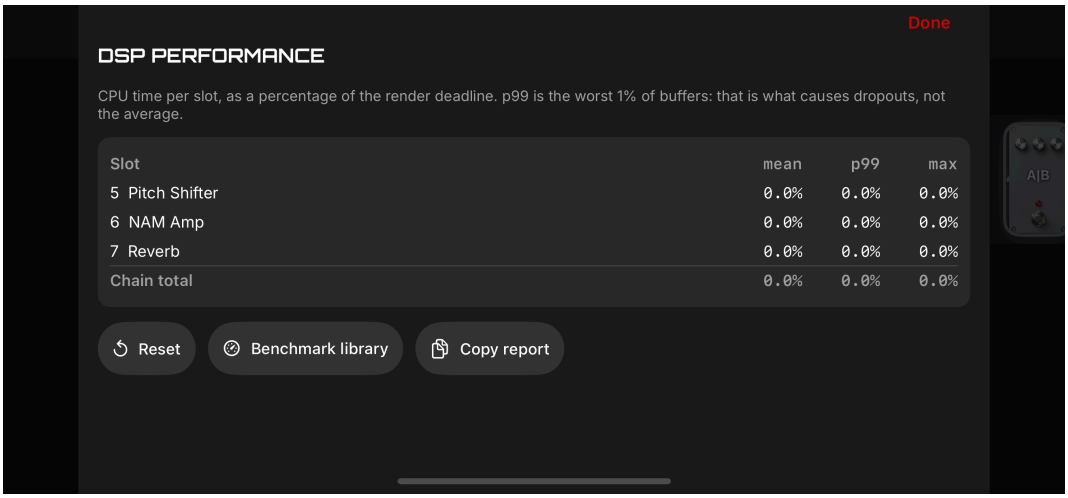

Figure 4: DSP Performance panel — per-pedal cost and the model benchmark

2.1.5 Scene Row

Eight scene capsules sit under the meters when the screen has room (on short phones they fold into an “S” menu in the toolbar). Tap a capsule to switch scenes, long-press it to rename. See the **Scenes** chapter.

2.2 Row 2: Effects Chain Strip

The effects chain strip displays the 16 effect slots as a horizontal row of pedals between the **IN** tab on the left and the **OUT** tab on the right. On a phone the row scrolls sideways once it is wider than the screen. Each slot shows:

- **Effect icon** — The SF Symbol for the effect type
- **Short name** — Abbreviated effect name (e.g., “RVB”, “DLY”, “CMP”)
- **Category color** — A distinctive background tint identifying the effect category

2.2.1 Effect Category Colors

Category	Color	Effect Types
Dynamics	Blue	Noise Gate, Compressor, Limiter
Distortion/Overdrive	Orange	Screamer, Klear, Mouse, Fuzz 32, Turbo
Amplifier	Red	NAM Amp
Modulation	Purple	Chorus, Flanger, Phaser, Tremolo, Vibrato
Pitch Shift	Green	Poly Shifter
Wah/Synth	Cyan	Auto Wah, LFO Wah
EQ/Filters	Yellow	7-Band EQ, High Pass, Low Pass
Reverb	Teal	Room, Hall, Plate, Spring, Z-Verb
Delay	Mint	Digital, Tape Echo, BBD
Cabinet/IR	Brown	IR Loader

2.2.2 Slot Interactions

- **Tap** a slot to select it and view its parameters in Row 4
- **Double-tap** an empty slot to open the Effect Browser and add a new effect
- **Flick up or down** on a pedal to bypass or re-enable it
- **Long-press and drag** to reorder effects in the chain; while you drag a pedal in a series chain, a **fork strip** appears over the parameter area, and dropping onto a pedal there runs the two in parallel (see the **Signal Path** chapter)

- A bypassed effect shows a dimmed pedal with its LED off

2.2.3 IN and OUT Tabs

- **IN** opens the **Chain Overview**: the whole rig at full size with the Signal Path templates above it
- **OUT** shows the chain's output width, **mono** or **stereo**, so you can see at a glance when a stereo effect or a dual mono split is feeding both channels

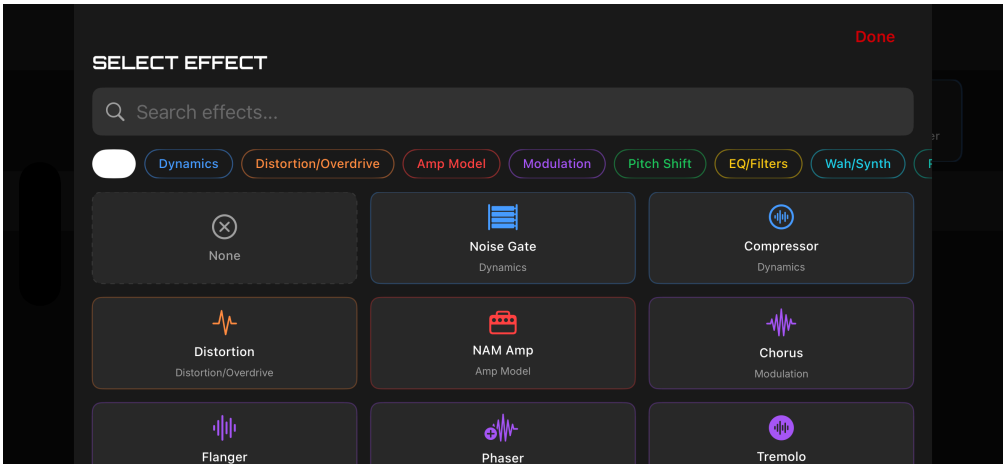

Figure 5: Effect Browser — double-tap a slot to add an effect

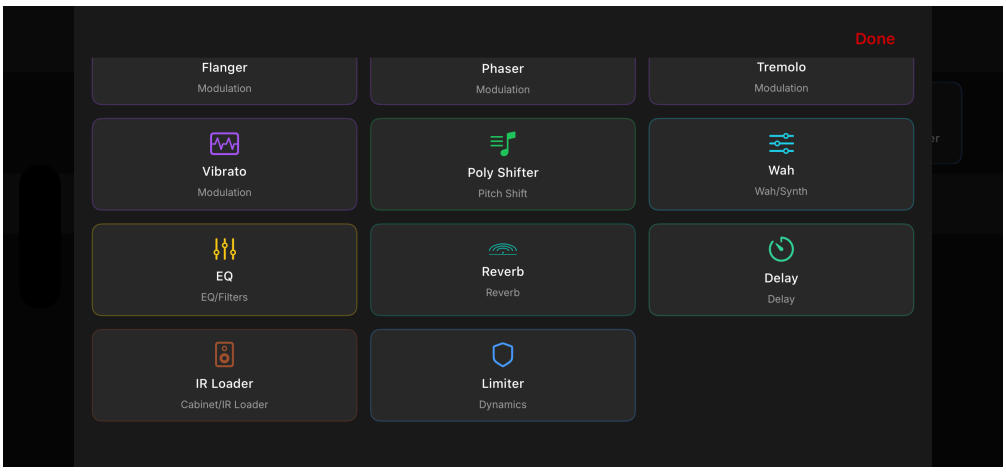

Figure 6: Effect Browser — browsing effect categories

2.3 Row 3: Algorithm Bar

When an effect that supports multiple algorithms is selected, the algorithm bar appears between the chain strip and parameter area. It shows the currently selected algorithm name (e.g., "Plate" for reverb, "Screamer" for distortion). Tap it to open the Algorithm Picker overlay, which lists all available algorithms with descriptions.

2.4 Row 4: Parameter Area

The parameter area fills the remaining screen space and displays controls for the currently selected effect:

- **Knob grid** — Rotary knobs for continuous parameters (drive, mix, decay, etc.)
- **Toggle switches** — For boolean parameters (e.g., tempo sync on/off)
- **Slider controls** — For parameters that benefit from linear input
- **Subdivision picker** — For tempo-synced effects, lets you choose note divisions

Each knob displays its parameter name, current value, and unit. **Double-tap** any knob or slider to instantly reset it to its midpoint value — useful for quickly returning a parameter to a neutral starting point.

3 Signal Path: Split Lanes & the Mix Block

3.1 Overview

A rig does not have to be one straight line. Ampz can split the chain once into two parallel lanes, **A** and **B**, and merge them again through a grey **Mix** pedal. Anything before the split runs in series into both lanes; anything after the merge runs in series on the sum. That is enough for a dual-amp rig (two NAM amps side by side), a wet/dry rig (effects on one lane, the dry guitar on the other), or a dual mono rig with lane A in the left speaker and lane B in the right.

The split is part of the preset and is saved with it. Scenes can change the Mix levels, pans and crossfade, but never the structure.

Figure 7: A split rig in the strip — lanes A and B as stacked rows between the Split node and the Mix pedal

3.2 Chain Overview

Tap the **IN** tab at the left end of the strip (or the branch icon on the Mix panel) to open the Chain Overview. It shows every pedal at full size, the lanes as two rows with their **A** and **B** labels, the Split node, the Mix pedal, and the **OUT** width tag. Tap a pedal to select it, tap an empty slot to open the Effect Browser, and tap **Close** to return.

Figure 8: Chain Overview — a series rig; Straight is highlighted in the Signal Path row

3.3 Signal Path Templates

The **Signal Path** row above the board lays the rig out in one tap. Applying a template never deletes or reorders your pedals: it moves the split and merge points and fills the new regions with empty outlined slots. The template the rig is currently in is highlighted; tapping it again does nothing and answers with a soft buzz instead of a click.

Template	Shape	Use it for
Straight	One row of 16	A classic series pedalboard
Middle Split	4, then 4 + 4, then 4	Pedals in front, two amps or two effect lanes in the middle, cab and limiter after the merge
Immediate Split	6 + 6 from the input, then 4	Two complete rigs from the guitar onward, summed to one output; pans are centred
Dual Mono	As Immediate Split, lane A left and lane B right	Two amps in stereo: A in the left speaker, B in the right

Figure 9: Middle Split — four pedals, two lanes of four, then four

Figure 10: Dual Mono — the lanes take the lane colours, the Mix pedal reads A|B, and L and R tags mark the sides

A few rules worth knowing:

- On a rig with more pedals than a template's region has room for, the regions grow to fit and the empty slots shrink. A template that cannot fit two lanes at all leaves the rig in series.
- A series rig fills a template's first region and spills the rest after the merge, so your cab and limiter stay last.
- **Straight** remembers where the lanes were. Tap Straight, then tap a split template again, and the same pedals go back into the same lanes. The memory lasts until you move or change a pedal, load a preset, or split by hand.

- **Dual Mono** writes its pans into every scene of the preset, so switching scenes keeps the stereo image.

3.4 Splitting by Hand

- **Split** (under the overview board) needs a selected pedal in a series chain and at least one free slot. It puts the selected pedal in lane A and a new empty slot in lane B.
- **Drag a pedal onto another pedal** in the fork strip that appears while you drag in a series chain, and the two run in parallel.
- **Drag the Split node onto a pedal** to move the split there; **drag the Mix pedal onto a pedal** to move the merge point. With VoiceOver, the Split node and the Mix pedal carry **Move split left/right** and **Move merge left/right** actions.
- The dashed **+** at the end of a lane adds an empty slot to that lane when the rig has one spare.
- **Stereo A|B** pans lane A hard left and lane B hard right with the crossfade centred. **Straight** (in the Signal Path row or on the Mix panel) returns to series.

3.5 The Mix Block

Tap the grey **Mix** pedal to open its panel.

Figure 11: Mix block — lane levels and pans, Crossfade, Delay trim, and the polarity and layout buttons

Control	Range	What it does
Level A / Level B	–60…+12 dB	Each lane's level into the merge
Pan A / Pan B	L…R	Each lane's position in the stereo field. Pan is a balance with unity at centre, so a split with one lane muted sounds identical to series

Control	Range	What it does
Crossfade	A…B	Centre plays both lanes at their levels; the ends mute the other lane. This is the control the Live **PATH** pad and the MIDI path actions move
Delay	–30…+30 ms	Trim on top of the automatic lane alignment, for taste or for a slapback-style offset
Ø B	on / off	Inverts lane B's polarity, for lanes that cancel when summed

Ampz measures the latency of each lane's pedals (IR loaders and pitch shifters add some) and delays the shorter lane so the two arrive together. The panel shows the alignment it applied; the **Delay** knob adds to it.

3.6 Lane Tags

- **A** and **B** label the lane rows; **B Ø** means lane B is inverted.
- A lane the crossfade has silenced dims as a row and shows a mute glyph. Its pedals still work and still show their own LEDs, because that is what the lane will sound like when you bring it back.
- **L** and **R** after a lane mean the rig is dual mono: that lane is panned hard to that side. The Mix pedal reads **A|B** instead of **MIX**.
- **DRY** after an empty lane means that lane passes your signal through unchanged into the mix. A split with pedals on one side only is therefore a wet/dry blend; move the Crossfade toward the pedal lane if you want only that lane. Two empty lanes simply pass the signal through.

TIP: For a dual-amp rig: tap **Middle Split** or **Immediate Split**, put a NAM amp and an IR in each lane, then tap **Dual Mono** for stereo or leave the pans centred for a summed mono blend. Watch the DSP badge: every full-size amp capture costs CPU, and the Quality knob on each NAM amp buys headroom.

4 NAM Amplifier

4.1 What is Neural Amp Modeling?

Neural Amp Modeling (NAM) uses deep neural networks to capture the sonic behavior of real guitar amplifiers. During a training process, a reference amplifier is measured at thousands of signal levels and frequencies. The resulting neural network (using WaveNet or LSTM architectures) can then reproduce the amplifier's response in real time.

Ampz runs the NAM inference engine in C++ using the NeuralAudio library with Eigen linear algebra and Apple's Accelerate framework for SIMD optimization. The result is authentic amplifier tone with latency under 2.5 ms.

4.2 Getting NAM Models

Ampz does not ship with bundled NAM models — you bring your own. There are two ways to get models:

- **Tone3000** — Browse and download thousands of community-created NAM models directly in-app. Sign in with a free Tone3000 account, search by amp name, brand, or genre, and download with one tap.
- **Import from device** — If you already have `.nam` files (from ToneHunt, NAM community forums, or your own captures), import them via the iOS Files share sheet or "Open In" menu.

TIP: New to NAM? Start with the Tone3000 tab — search for amp styles like "clean twin", "high-gain crunch", or "classic British" to find community models that match your playing style.

4.3 Model Browser

Tap the NAM Amplifier slot to view its controls. The model browser provides two tabs (placed next to the Done button at the top of the sheet for quick switching):

- **My Models** — Lists all `.nam` files stored locally on your device. Each row shows a cover thumbnail (auto-fetched from Tone3000 when available) plus tag chips, so you can scan your library visually. Import models by sharing `.nam` files to Ampz from the Files app or other sources.
- **Tone3000** — Browse and download community-created NAM models from the Tone3000 online platform. Requires authentication (free account).

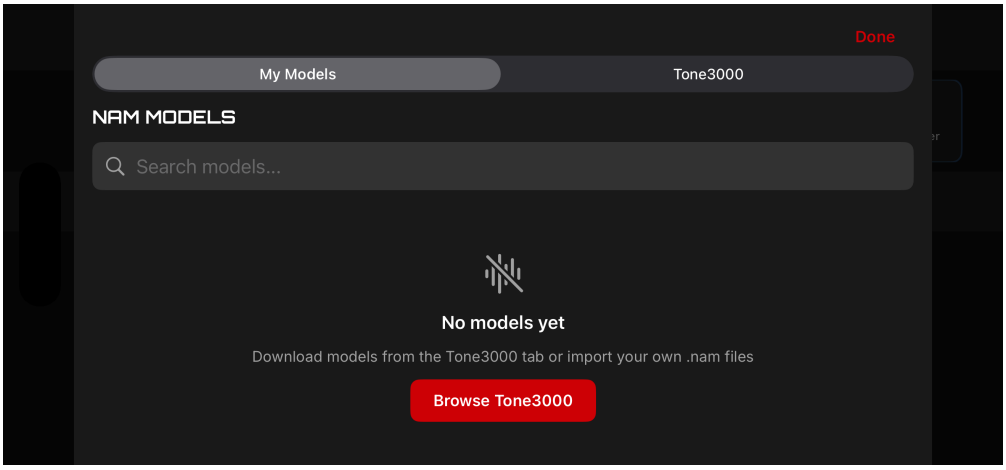

Figure 12: Model Browser — empty model list before importing

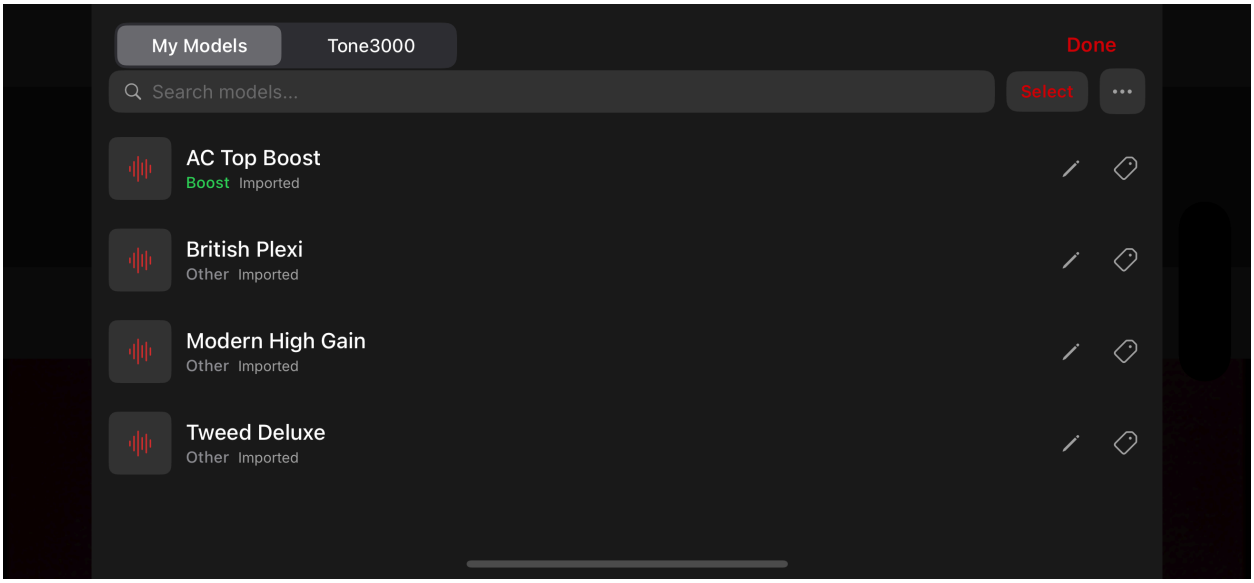

Figure 13: My Models — cover thumbnails, tag chips, and Select for bulk operations

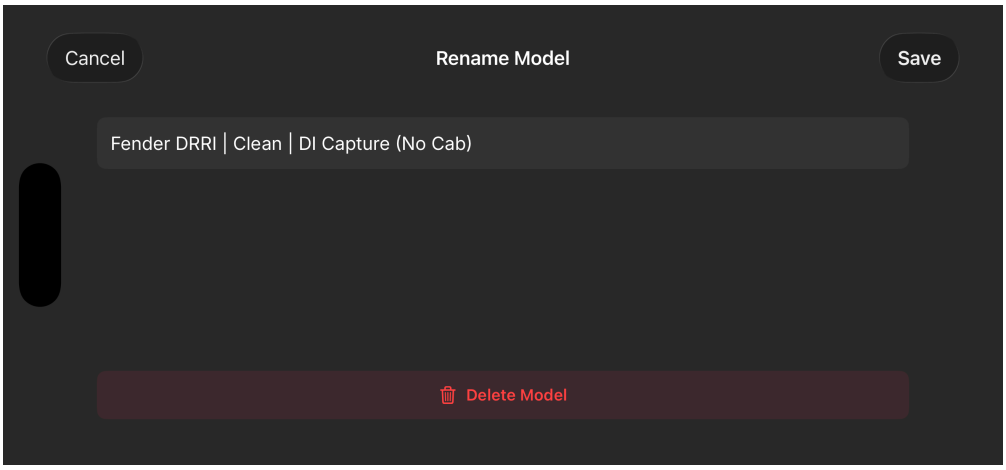

Figure 14: Renaming a NAM model

4.3.1 Tagging & Filtering Models

Organize your models with custom tags for quick filtering:

1. Tap the **tag icon** next to any model in the list
2. In the tag sheet, type a tag name (e.g., “crunch”, “recording”, “live”) and tap **Add**
3. Remove tags by tapping the × on any tag chip
4. Tap **Done** to save

When models have tags assigned, a row of filter chips appears below the search bar. Tap a tag chip to show only models with that tag. Tap **All** to clear the filter. Tags also work with search — typing a tag name in the search field will match models with that tag even if the model name doesn’t contain the search term.

TIP: Use tags to organize models by use case (e.g., “live”, “studio”), genre (e.g., “blues”, “metal”), or amp type (e.g., “Fender”, “Marshall”) for quick recall during sessions.

4.3.2 Deleting Models

To delete a single model, tap the pencil icon next to it, then tap **Delete Model** at the bottom of the rename sheet and confirm.

To delete multiple models at once, use multi-select mode:

1. Tap **Select** in the top-right corner of the My Models tab
2. Tap each model you want to remove — selected models show a filled red checkbox
3. Bundled (factory) models cannot be selected
4. Use **Select All** at the bottom-left to select every imported model
5. Tap the red **Delete (N)** button at the bottom-right
6. Confirm the deletion in the alert dialog

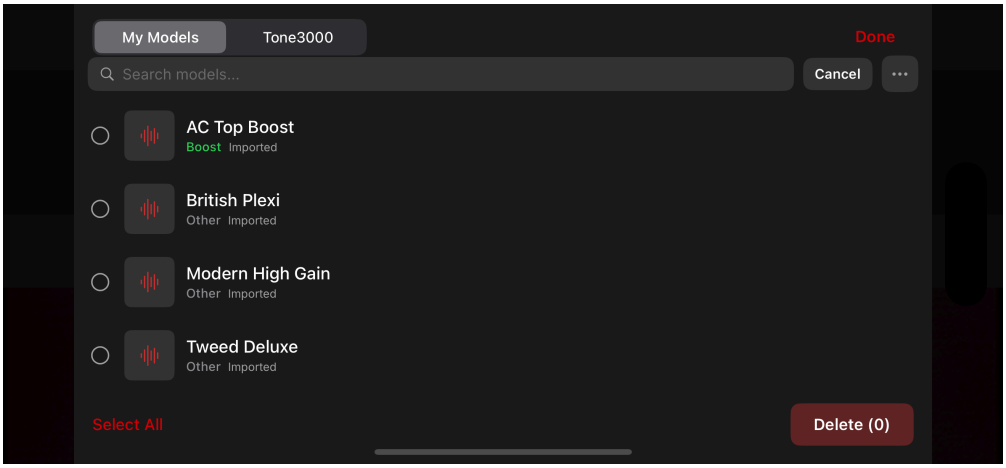

Figure 15: Multi-select mode — tap Select to enter, then choose models to delete

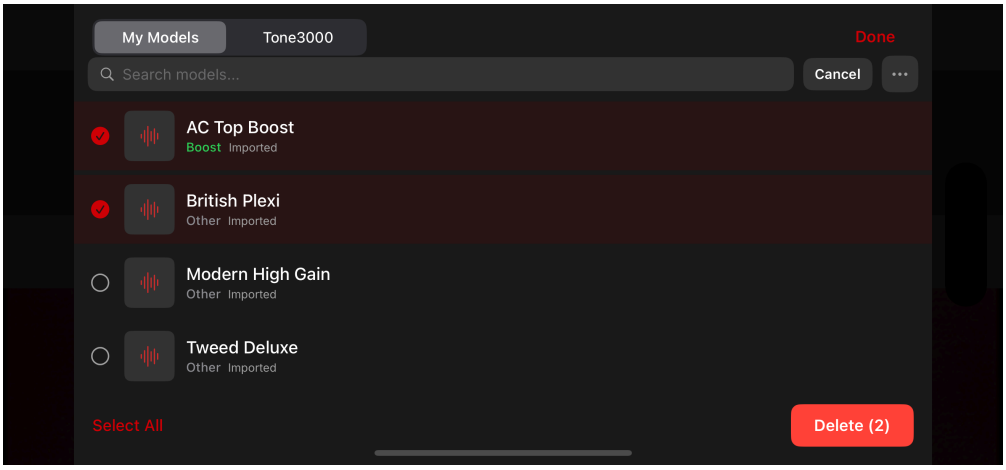

Figure 16: Selected models highlighted with red checkboxes and Delete button visible

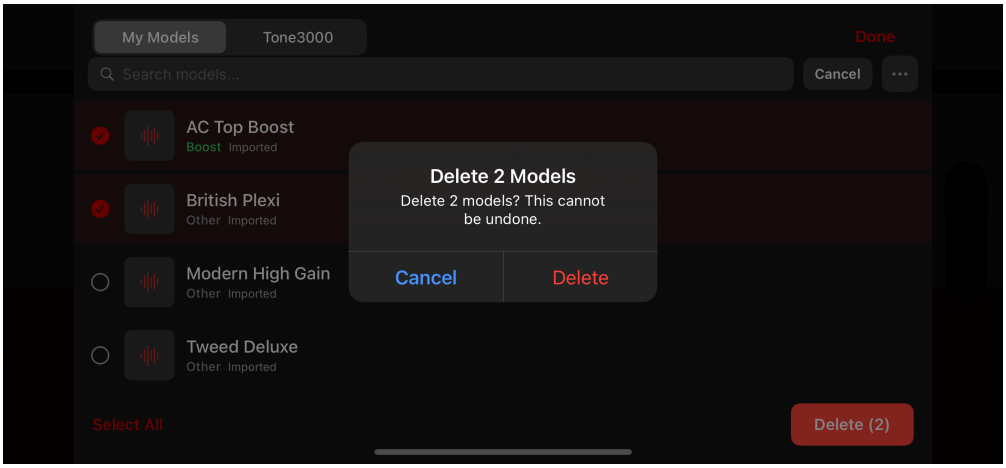

Figure 17: Confirmation alert before batch deletion

NOTE: If the currently loaded NAM model is among the selected models, Ampz will automatically unload it from the audio engine before deleting.

4.3.3 Importing .nam Files

1. Open the Files app or a file-sharing service
2. Locate the `.nam` file
3. Tap Share → Ampz, or use “Open In” → Ampz
4. The model appears in “My Models” and can be loaded immediately

4.3.4 Tone3000 Integration

Tone3000 is a community platform for sharing NAM models. The first time you tap the Tone3000 tab you’ll be prompted to sign in (or create a free account); from then on the browser opens directly to the model list.

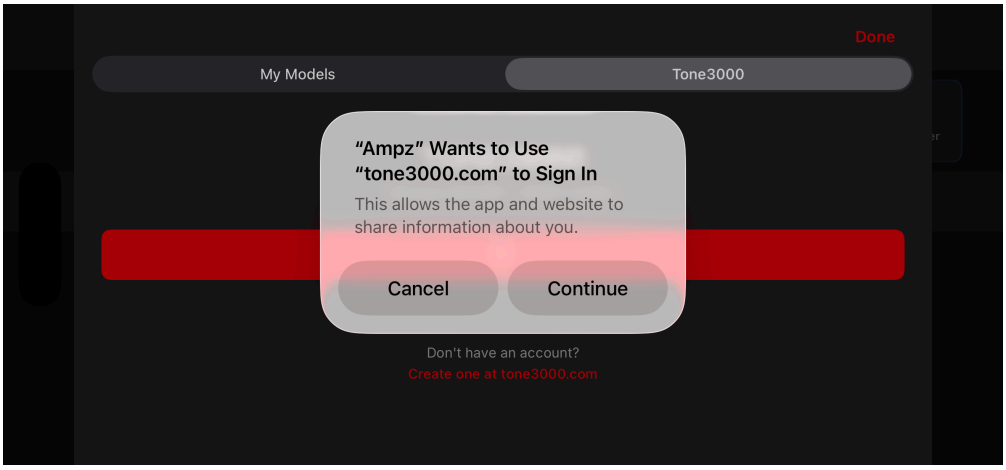

Figure 18: Tone3000 — initial redirect to sign in

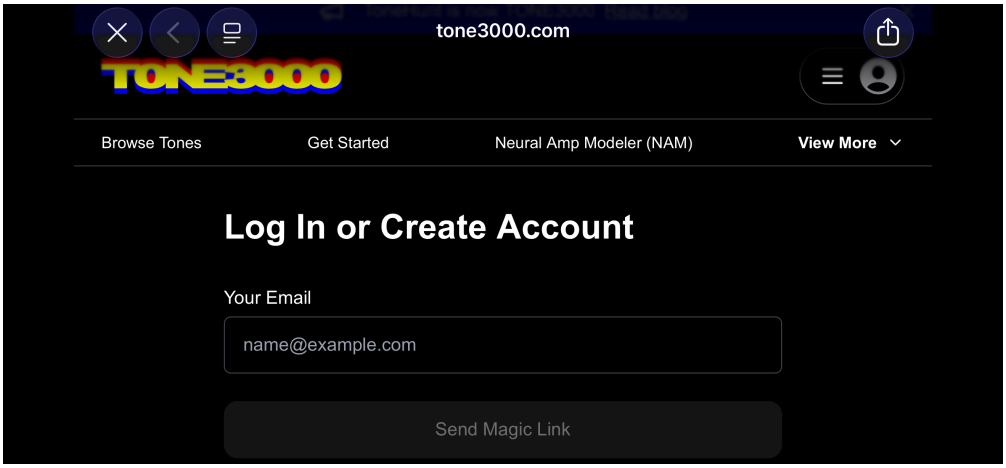

Figure 19: Tone3000 — website authorization

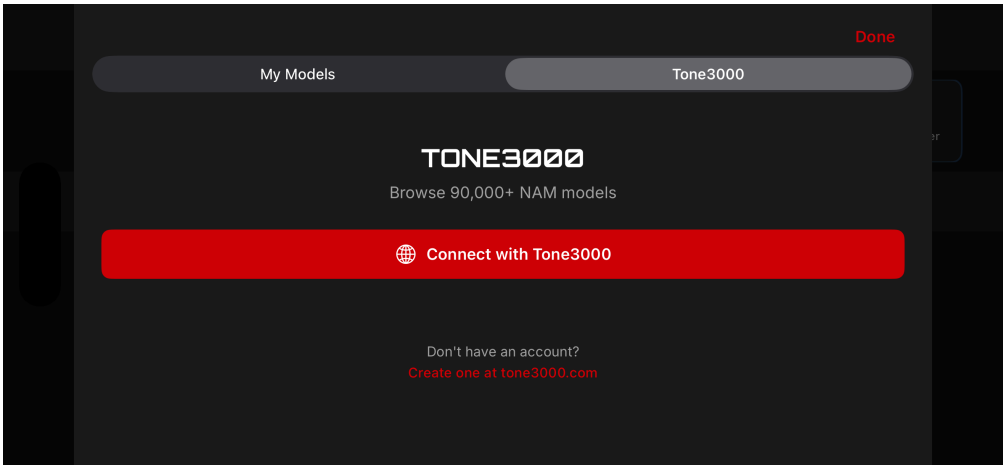

Figure 20: Tone3000 — login screen

4.3.4.1 Browsing

Each model in the list is shown as a **rich card** with a cover image, the model title, the capture count, the contributor's handle, and download / favorite stats. Two segmented controls at the top — **Browse** / **Favorites** — let you switch between the global library and your favourited captures. A row of category chips (Amps, Full Rigs, Pedals, Outboard) narrows the list further.

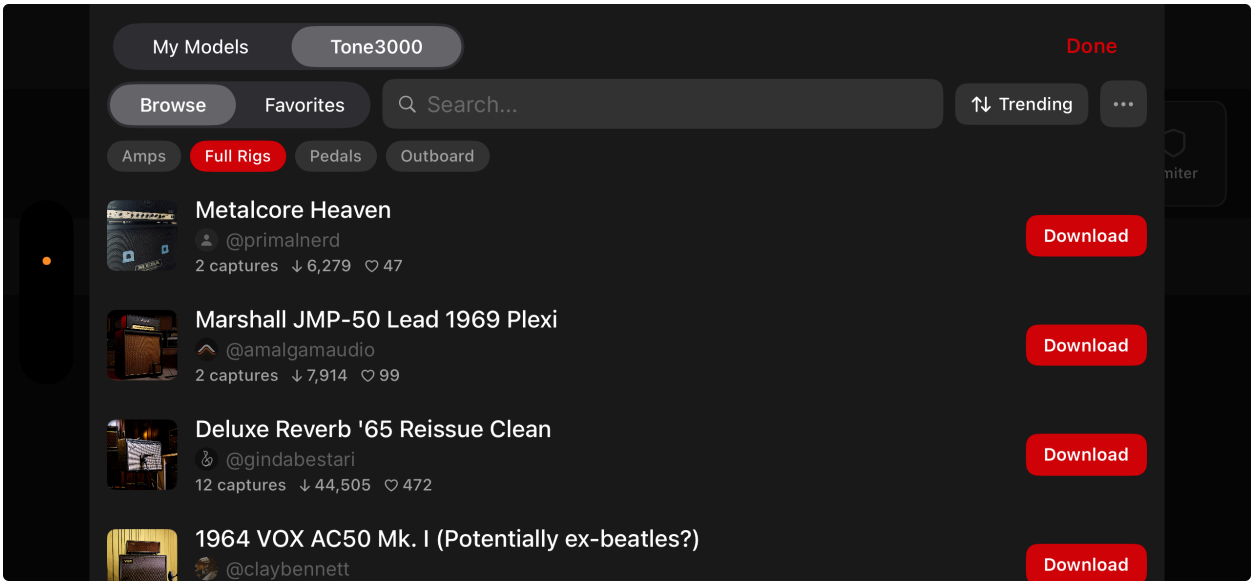

Figure 21: Tone3000 NAM list — rich cards with cover images and download stats

4.3.4.2 Sorting

Tap the sort menu (the chevron-up-down icon in the toolbar) to change ordering. Options: **Trending**, **Newest**, **Oldest**, **Most Downloaded**.

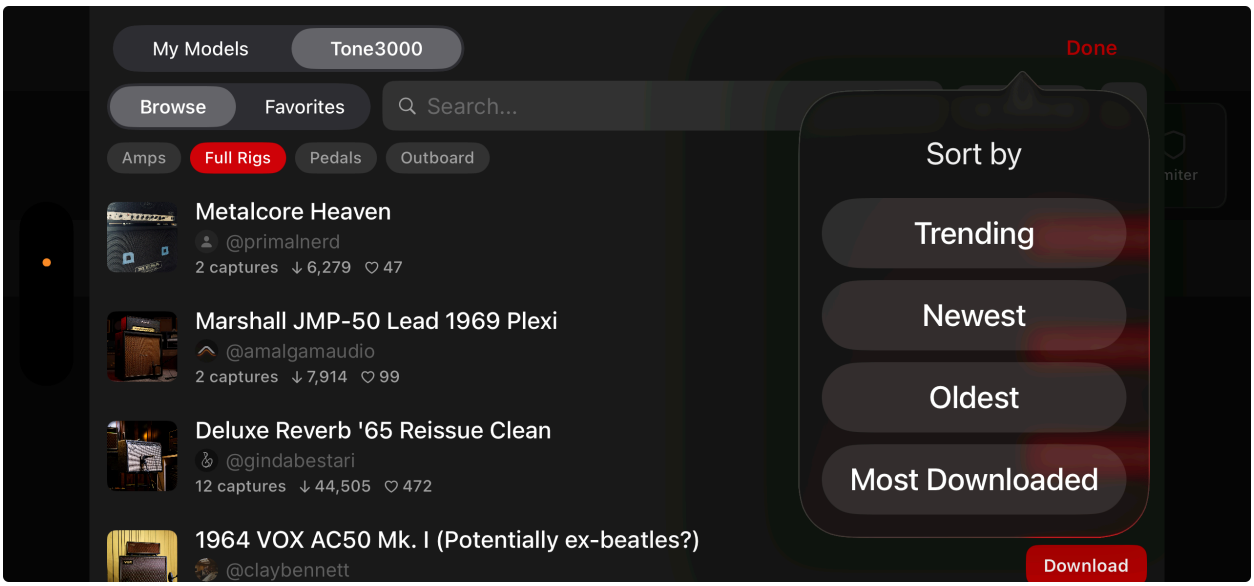

Figure 22: Sort menu — Trending, Newest, Oldest, Most Downloaded

4.3.4.3 Model Detail & Download

Tap any card to open its **detail view** — a full-bleed cover image, model title, contributor info, and a **Choose Files** button. Tap **Choose Files** to see the full list of NAM files attached to the capture (a single model can include multiple variants — different cabs, mic positions, room/no-room takes, etc.).

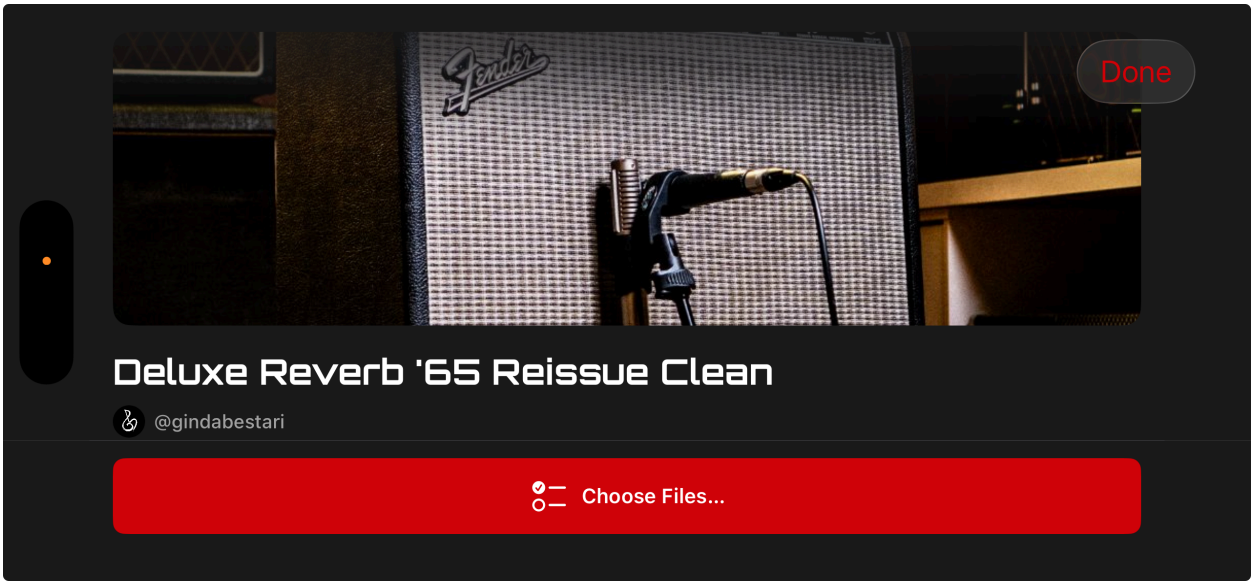

Figure 23: NAM detail view — cover image, title, contributor, and Choose Files

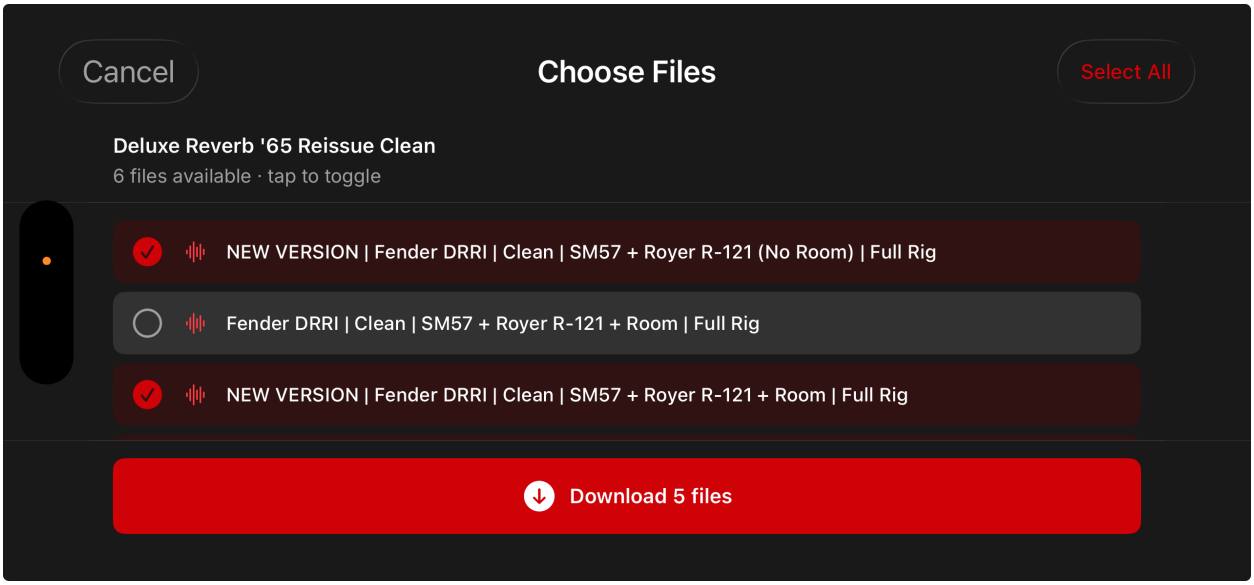

Figure 24: Choose Files — toggle individual variants on or off before downloading

Select the variants you want with the checkmarks (or use **Select All**), then tap the **Download N files** button. Downloads land in your **My Models** tab automatically, with cover images pre-populated.

TIP: New to NAM? Start with the **Full Rigs** chip on Trending sort — full rigs include the amp, cabinet IR, and microphone capture in a single file, so they sound great out of the box without a separate IR loader.

4.4 NAM Controls

When a NAM model is loaded, the parameter area shows the amp's rotary knobs, and the algorithm bar (above the knobs) provides the output-mode selector plus the Gate, EQ, and Bypass toggles.

NOTE: Any slot can be a NAM amp. The red **Amplifier** slot in the default rig is only the starting point — any of the 16 chain slots can load its own `.nam` model, so you can run several captures at once (for example a clean amp feeding a boosted lead amp).

Knobs:

Knob	Range	Description
INPUT	–24…+24 dB	Input level into the NAM model
GATE	–100…0 dB	Noise-gate threshold applied before the model — reduces hum and hiss
BASS	0–10	Low-frequency EQ applied after NAM processing
MIDDLE	0–10	Mid-frequency EQ applied after NAM processing
TREBLE	0–10	High-frequency EQ applied after NAM processing
OUTPUT	–40…+40 dB	Output level after all NAM processing
QUALITY	0–100%	CPU / fidelity trade-off — **A2 slimmable models only**; lower values run a smaller sub-model for less CPU

The **QUALITY** knob appears only when the loaded model is an A2 slimmable capture; non-slimmable models show the six standard knobs.

Algorithm Bar:

Control	Default	Description
Output mode	Normalized	Cycles **Raw** → **Normalized** → **Calibrated**. Raw passes the model output untouched; Normalized levels every model to a consistent loudness; Calibrated matches real-world gain staging
Gate	On	Enables/disables the input noise gate
EQ	On	Enables/disables the 3-band EQ (Bass, Middle, Treble)
Bypass	Off	Bypasses the entire NAM amplifier slot

In **Calibrated** output mode, two extra controls appear: an **Input Cal** knob (–15…+24 dBu — set to your audio interface's reference level) and a **Cal In** toggle.

Figure 25: NAM Amplifier — knobs and algorithm bar toggles

Figure 26: NAM Amplifier — with Gate and EQ toggled off

5 Effects Reference

Ampz provides 17 effect types organized into 10 categories. Each effect can be loaded into any of the 16 chain slots and supports multiple algorithm variants.

5.1 Dynamics

5.1.1 Noise Gate

Removes unwanted noise (hum, buzz) when you're not playing. The gate opens when the input signal exceeds the threshold and closes when it drops below.

Parameter	Range	Default	Description
Threshold	–80 to 0 dB	–40 dB	Signal level below which the gate closes
Attack	0.1–50 ms	1 ms	How quickly the gate opens
Release	5–500 ms	50 ms	How quickly the gate closes after signal drops
Range	0–96 dB	80 dB	Maximum attenuation when gate is closed

5.1.2 Compressor

Reduces the dynamic range of your signal by attenuating loud peaks and boosting quiet passages. Essential for even sustain and consistent volume.

Parameter	Range	Default	Description
Threshold	–60 to 0 dB	–20 dB	Level above which compression begins
Ratio	1:1–20:1	4:1	Compression ratio (higher = more compression)
Attack	0.1–100 ms	10 ms	How quickly compression engages
Release	10–1000 ms	100 ms	How quickly compression releases
Gain	0–30 dB	0 dB	Makeup gain to compensate for volume reduction

5.1.3 Limiter

A hard ceiling on output level to prevent clipping and protect speakers. Acts as a compressor with an infinite ratio.

Parameter	Range	Default	Description
Threshold	–30 to 0 dB	–1 dB	Maximum output level allowed
Release	10–500 ms	50 ms	How quickly the limiter releases

TIP: Place the limiter at the end of your effects chain (slot 9) to protect your output from unexpected volume spikes.

Figure 27: Limiter parameters

5.2 Distortion / Overdrive

All five distortion algorithms use Wave Digital Filter (WDF) circuit modeling for authentic analog behavior.

5.2.1 Screamer

Mid-hump overdrive with WDF diode clipping. Inspired by classic green pedal circuits — warm, smooth, and musical.

Parameter	Range	Default	Description
Drive	0–100%	40%	Amount of overdrive / gain
Tone	0–100%	60%	Tilt EQ — lower values are darker, higher values are brighter
Level	0–100%	70%	Output volume

Parameter	Range	Default	Description
Mix	0–100%	100%	Dry/wet blend

5.2.2 Klear

Transparent overdrive with germanium diodes and an inverse-ganged clean blend. Preserves your original tone while adding harmonic richness.

Parameter	Range	Default	Description
Drive	0–100%	50%	Overdrive intensity
Tone	0–100%	50%	Brightness control
Level	0–100%	70%	Output volume
Mix	0–100%	100%	Dry/wet blend

5.2.3 Mouse

Hard-clipping distortion with shunt diodes and LM308 slew limiting. Aggressive, wide-range distortion from subtle grit to full saturation.

Parameter	Range	Default	Description
Drive	0–100%	35%	Distortion amount
Tone	0–100%	50%	Brightness control
Level	0–100%	50%	Output volume
Mix	0–100%	100%	Dry/wet blend

5.2.4 Fuzz 32

Germanium-voiced fuzz with two-stage asymmetric clipping and soft limiting for classic fuzz sustain. Warm, compressed, and musical across the full drive range. Includes a built-in output noise gate to tame the high-gain noise floor without choking sustain.

Parameter	Range	Default	Description
Fuzz	0–100%	60%	Fuzz intensity and sustain
Tone	0–100%	50%	Brightness (500–4500 Hz sweep)
Level	0–100%	70%	Output volume
Gate	0–100%	50%	Output noise gate threshold — higher values reduce more noise
Mix	0–100%	100%	Dry/wet blend

Figure 28: Fuzz 32 — Fuzz, Tone, Level, Gate, and Mix controls

5.2.5 Turbo

LED-clipped distortion with LM308 slew limiting. Based on the same circuit topology as Mouse but with LED-style diodes for higher headroom — open, loud, and less compressed.

Parameter	Range	Default	Description
Drive	0–100%	35%	Distortion amount
Tone	0–100%	50%	Brightness control
Level	0–100%	50%	Output volume
Mix	0–100%	100%	Dry/wet blend

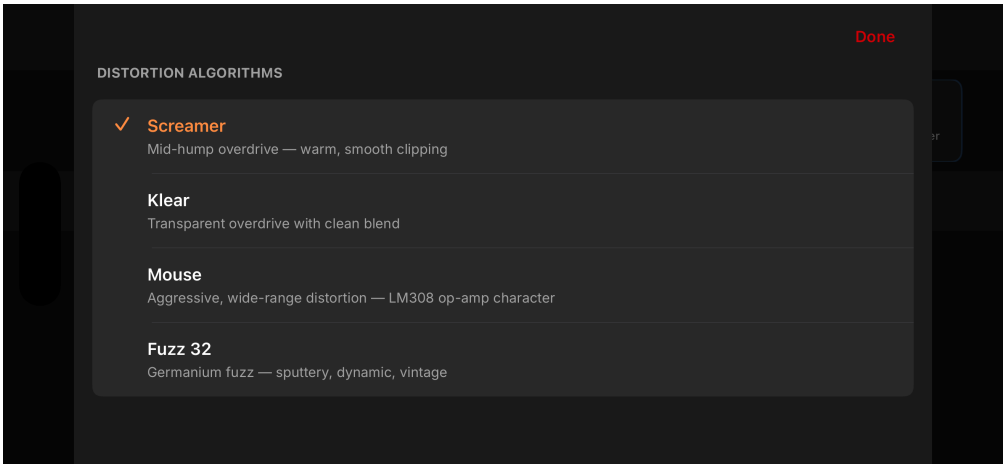

Figure 29: Distortion algorithm picker

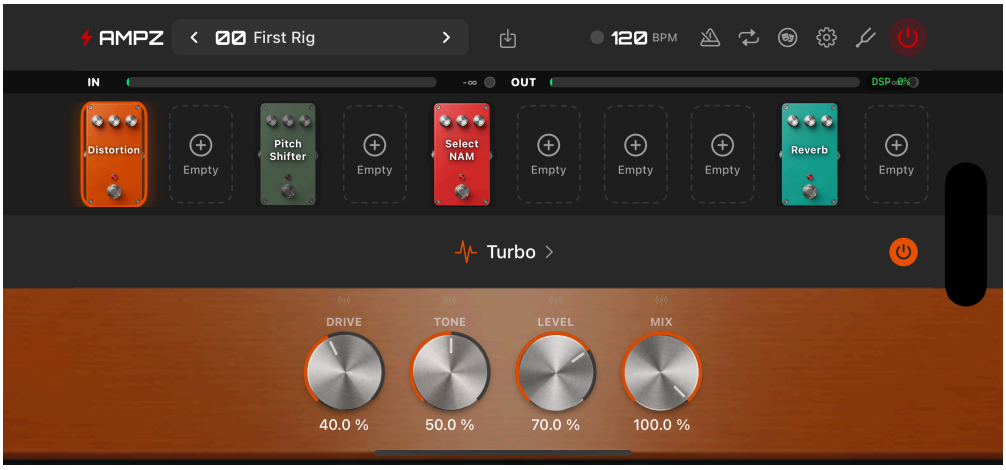

Figure 30: Turbo distortion — LED-clipped with high headroom

5.3 Modulation

5.3.1 Chorus

Adds width and shimmer by mixing the dry signal with pitch-modulated copies. Three algorithm variants:

- **Classic** — Traditional mono chorus with subtle shimmer
- **Dimension** — Wide stereo chorus inspired by classic 1980s rackmount units
- **Tri-Chorus** — Three-voice chorus for lush, thick, wide textures

Parameter	Range	Default	Description
Rate	0.05–10 Hz	0.8 Hz	LFO speed
Depth	0–100%	50%	Modulation depth
Mix	0–100%	50%	Dry/wet blend
Tone	0–100%	50%	High-frequency rolloff of wet signal

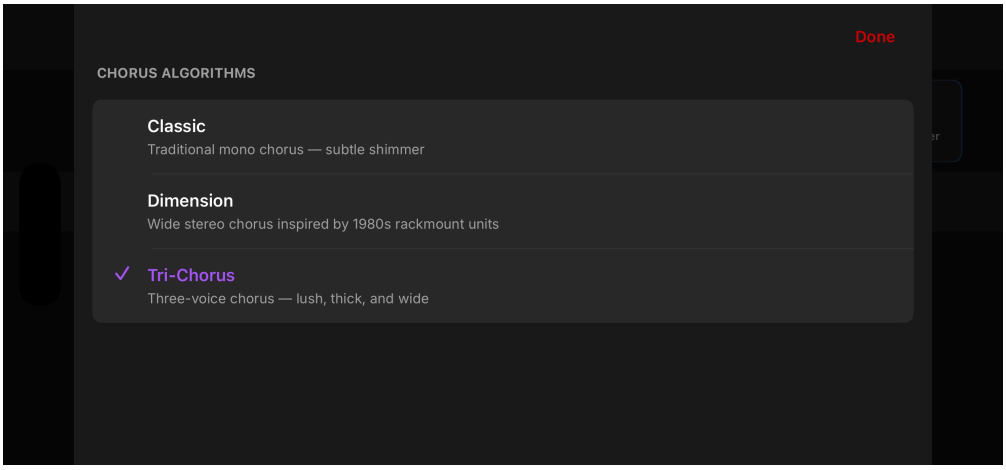

Figure 31: Chorus algorithm picker

5.3.2 Flanger

Stereo flanger with negative feedback for the classic “jet” sweep sound. Uses short modulated delay lines with phase-inverted feedback.

Parameter	Range	Default	Description
Rate	0.05–5 Hz	0.3 Hz	Sweep speed
Depth	0–100%	60%	Sweep range
Feedback	–100 to +100%	–50%	Resonance (negative = classic jet sound)
Mix	0–100%	50%	Dry/wet blend

5.3.3 Phaser

Multi-stage all-pass filter phaser with stereo sweep. Two algorithm variants:

- **4-Stage** — Smooth, subtle phase sweep (MXR Phase 90 style)
- **6-Stage** — Deeper, more pronounced phasing with richer notch patterns

Parameter	Range	Default	Description
Rate	0.05–5 Hz	0.5 Hz	Sweep speed
Depth	0–100%	70%	Modulation depth
Feedback	0–100%	50%	Resonance — higher values intensify the effect
Mix	0–100%	50%	Dry/wet blend

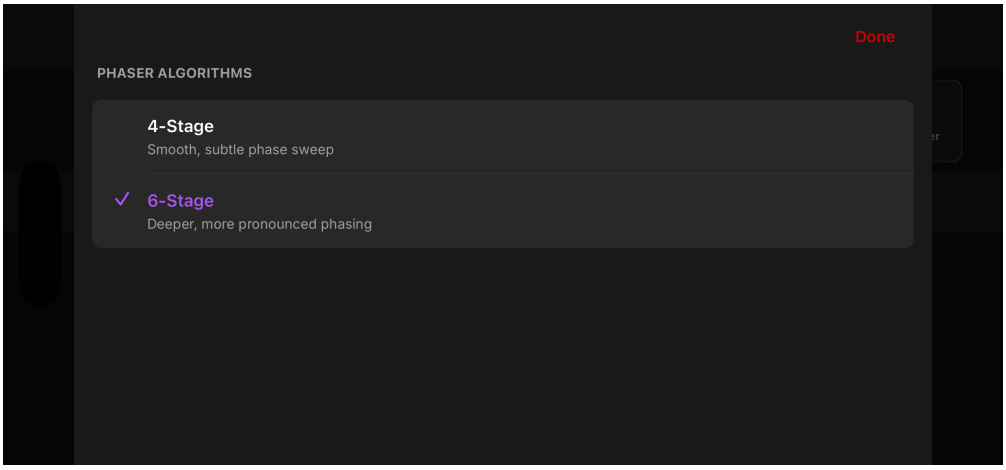

Figure 32: Phaser algorithm picker

5.3.4 Tremolo

Amplitude modulation — rhythmic volume variation. Two waveform shapes:

- **Sine** — Smooth, classic tremolo modulation
- **Triangle** — Sharper, more percussive volume pulsing

Parameter	Range	Default	Description
Frequency	0.5–20 Hz	4 Hz	Modulation rate
Depth	0–100%	50%	How deep the volume dips
Waveform	Sine / Triangle	Sine	Shape of the modulation wave

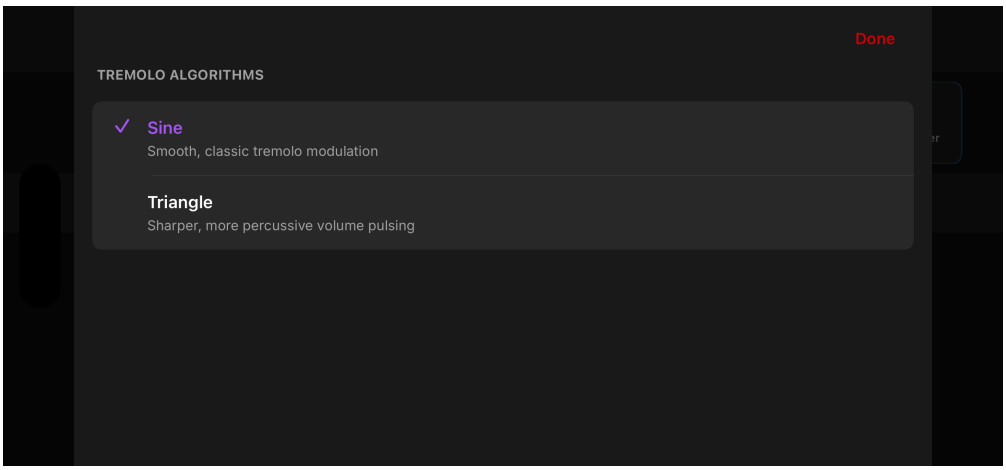

Figure 33: Tremolo algorithm picker

Figure 34: Tremolo effect parameters

5.3.5 Vibrato

Pure pitch modulation via a modulated delay line — 100% wet signal with no feedback. Two waveform shapes:

- **Sine** — Smooth pitch wobble
- **Triangle** — Angular pitch modulation

Parameter	Range	Default	Description
Rate	0.5–10 Hz	5 Hz	Modulation speed
Depth	0–100%	50%	Pitch variation amount
Waveform	Sine / Triangle	Sine	Modulation wave shape

5.4 Pitch

5.4.1 Poly Shifter

Polyphonic pitch shifting with two algorithm variants:

- **Pitch Shift** — Low-latency polyphonic pitch shifting for drop tuning, virtual capo and harmonies. Pick attacks come through within a few milliseconds and low strings track cleanly. Pitch shift engine adapted from the open-source TONE3000 plugin (MIT).
- **Analog Octave** — Rectifier-based octave up/down with raw, gritty analog character.

Parameter	Range	Default	Description
Semitones	−12 to +12	0	Pitch shift in semitones
Cents	−50 to +50	0	Fine-tune in cents
Mix	0–100%	50%	Dry/wet blend

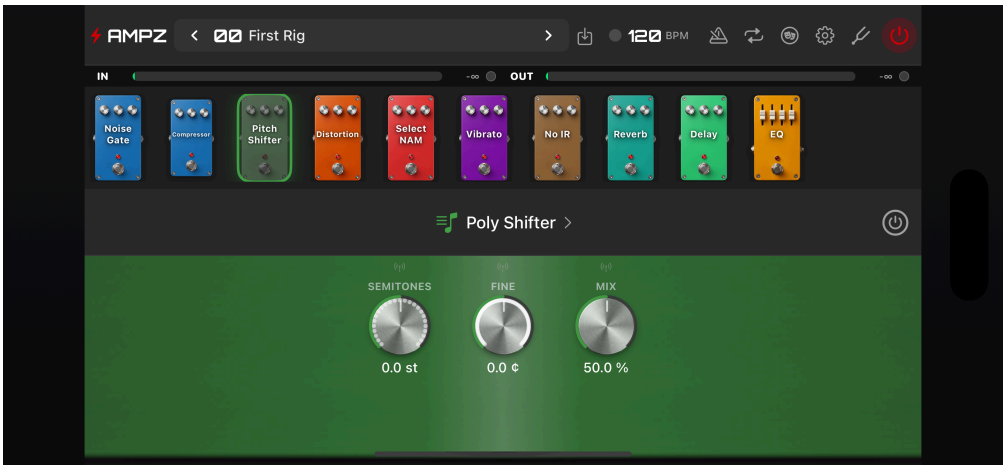

Figure 35: Poly Shifter algorithm options

5.5 Wah

5.5.1 LFO / Envelope

Two wah algorithms selectable from the algorithm picker:

- **LFO** — Cyclic wah sweep driven by an LFO oscillator. Rate and Depth control the sweep speed and range.
- **Envelope** — Touch-sensitive envelope filter (Mutron III style auto-wah). Responds to your playing dynamics — play harder for more wah sweep.

Toggle the **AUTO** button to switch between manual and automatic operation:

- **Auto OFF** — The Frequency knob directly controls the filter cutoff (manual wah).
- **Auto ON** — The selected algorithm (LFO or Envelope) sweeps the filter automatically. Additional parameters appear for the active algorithm.

Parameter	Range	Default	Description
Frequency	300–2800 Hz	800 Hz	Filter cutoff (manual) or sweep center (auto)
Q	1–20	5	Filter resonance — higher values give a more vocal, peaked sound
Rate	0.1–10 Hz	1 Hz	LFO sweep speed (LFO algorithm, Auto ON only)
Depth	0–100%	70%	Sweep range (Auto ON only)
Sensitivity	0–100%	50%	Envelope detection gain (Envelope algorithm, Auto ON only)
Attack	1–50 ms	10 ms	Envelope attack time (Envelope algorithm, Auto ON only)

Parameter	Range	Default	Description
Release	50–500 ms	200 ms	Envelope release time (Envelope algorithm, Auto ON only)
Direction	Up / Down	Up	Envelope sweep direction (Envelope algorithm, Auto ON only)
Mix	0–100%	100%	Dry/wet blend

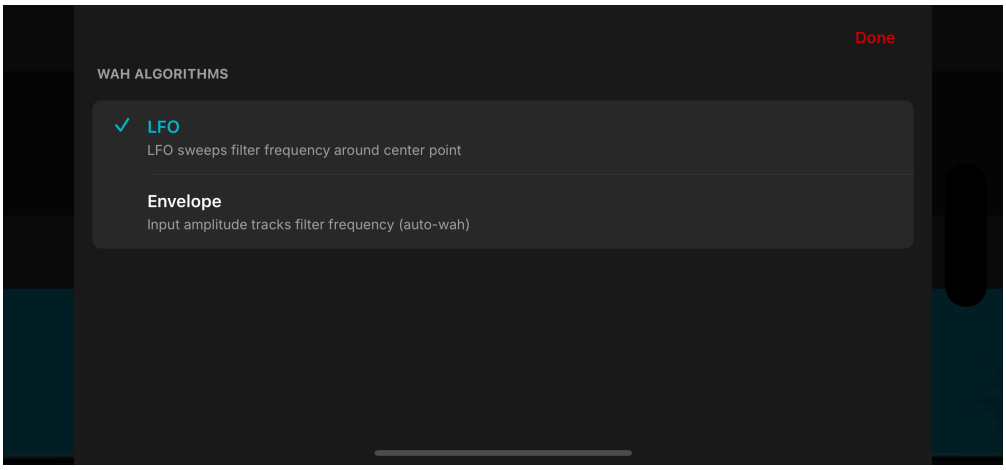

Figure 36: Wah algorithm picker — LFO and Envelope

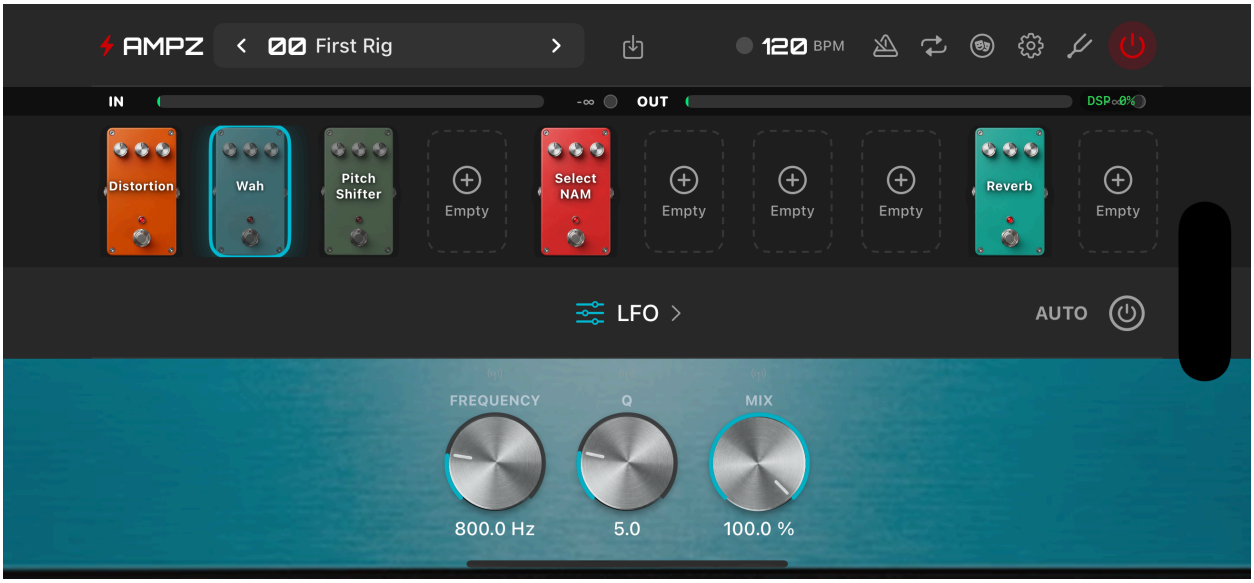

Figure 37: Wah with Auto OFF — manual frequency control

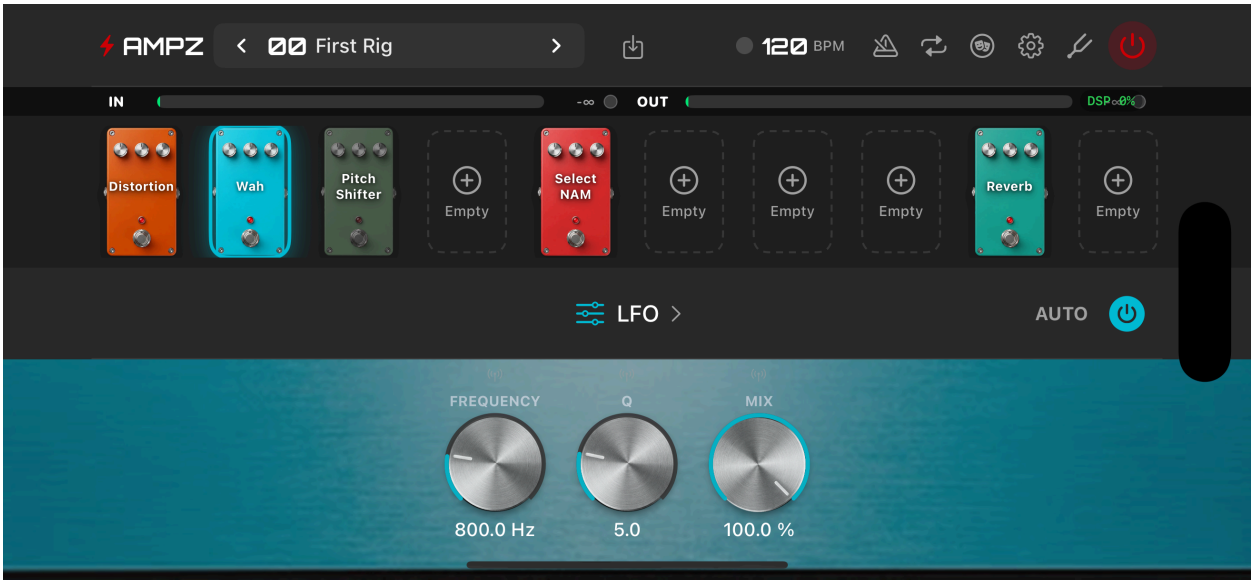

Figure 38: Wah with Auto ON — LFO Rate and Depth controls appear

5.6 EQ / Filters

5.6.1 7-Band EQ

Professional 7-band parametric equalizer with 5 parametric bands plus low and high shelving filters.

Band	Center Freq	Type
Band 1	80 Hz	Low shelf
Band 2	200 Hz	Parametric
Band 3	500 Hz	Parametric
Band 4	1 kHz	Parametric
Band 5	2 kHz	Parametric
Band 6	5 kHz	Parametric
Band 7	12 kHz	High shelf

Each band has a gain range of ±12 dB. An output level control compensates for overall volume changes.

5.6.2 High Pass Filter

Butterworth 2nd-order high-pass filter for removing low frequencies (rumble, subsonic content).

Parameter	Range	Default	Description
Frequency	20–2000 Hz	80 Hz	Cutoff frequency — frequencies below this are attenuated

Parameter	Range	Default	Description
Resonance	0.1–10	0.707	Q factor — higher values create a resonant peak at cutoff

5.6.3 Low Pass Filter

Butterworth 2nd-order low-pass filter for removing high frequencies (fizz, harshness).

Parameter	Range	Default	Description
Frequency	200–20000 Hz	8000 Hz	Cutoff frequency — frequencies above this are attenuated
Resonance	0.1–10	0.707	Q factor — higher values create a resonant peak at cutoff

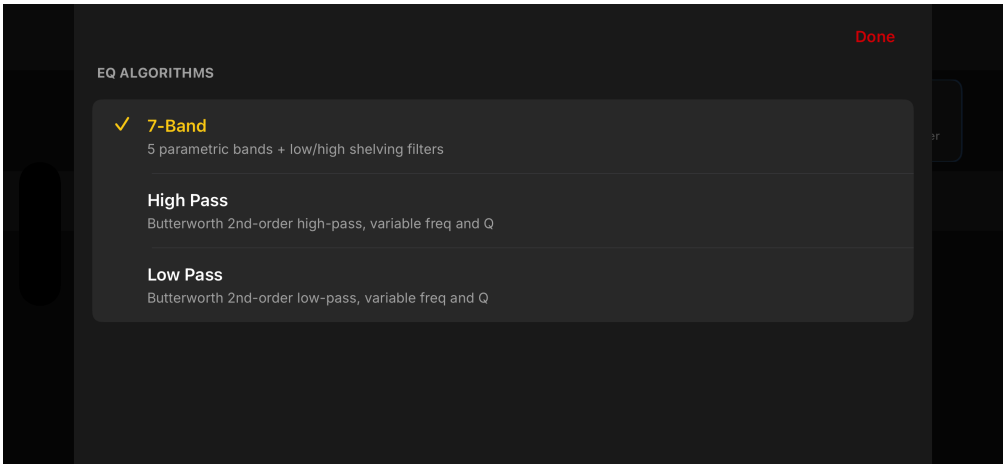

Figure 39: EQ algorithm picker

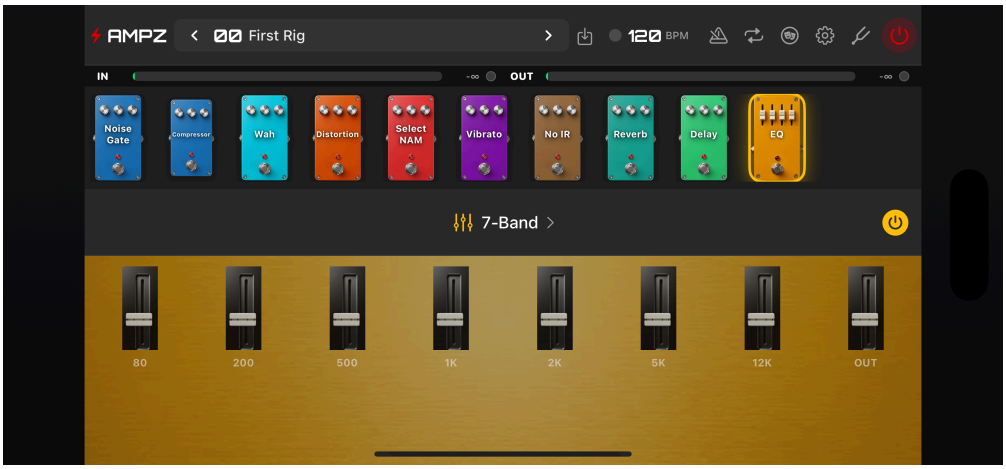

Figure 40: 7-Band EQ parameters

5.7 Reverb

Five reverb algorithms provide spaces ranging from tight rooms to expansive halls.

5.7.1 Room

Small reflective space with early reflections using the Freeverb algorithm. Tight and natural.

5.7.2 Hall

Large ambient space with long, evolving tails using the JPverb algorithm. Spacious and immersive.

5.7.3 Plate

Classic studio plate reverb — bright, dense, and lush. Uses the Dattorro plate algorithm for a smooth, metallic shimmer.

5.7.4 Spring

Physical spring tank model with transient-reactive dispersive chirp and a coupled ambient tail. Captures the characteristic drip, splash, and twang of vintage amp-built spring reverbs.

Pick the spring tank configuration with the **Type** control. Each type changes the diffusion and decay character — and a separate **Drip** knob controls the surf-style chirp that fires when you pick hard, on top of the ambient bed.

Spring-Specific Parameters:

Parameter	Range	Default	Description
Type	Short / Long / Tank	Long	Spring tank configuration — affects diffusion and decay character
Decay	0–100%	50%	Reverb tail length
Damping	0–100%	50%	High-frequency absorption — higher values darken the reverb
Crash	0–100%	100%	Transient crash sensitivity — simulates spring impact when picking hard
Drip	0–100%	0%	Surf-style dispersive chirp amount, layered on top of the ambient tail
Tail	0–100%	0%	Ambient bed level mixed into the reverb output
Mix	0–100%	40%	Dry/wet blend

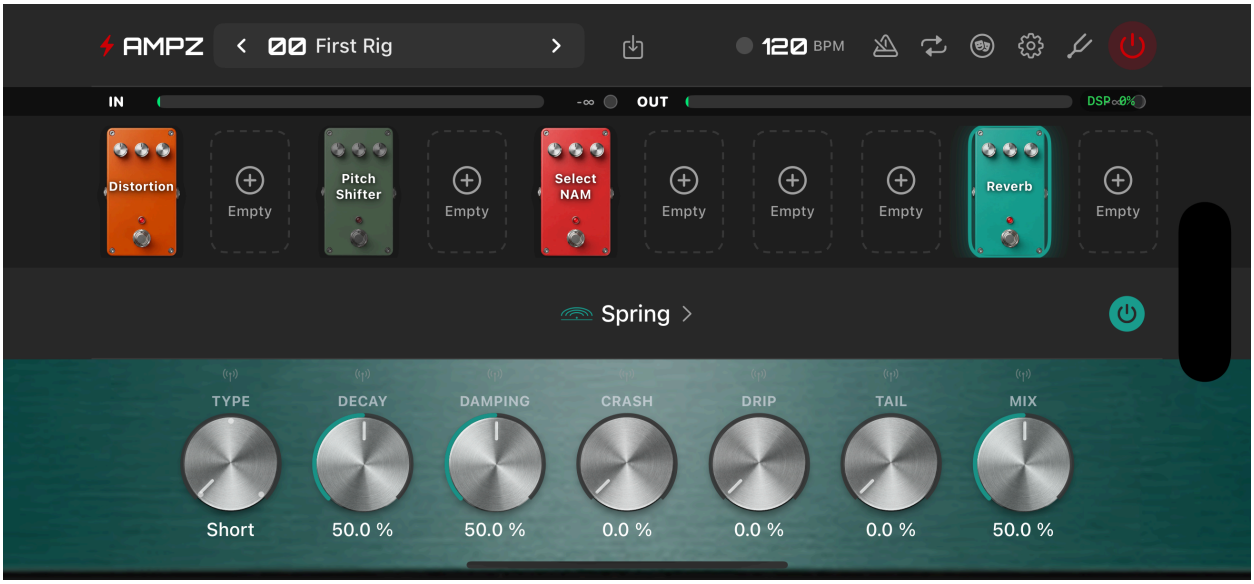

Figure 41: Spring reverb — Type, Decay, Damping, Crash, Drip, Tail, and Mix

5.7.5 Z-Verb

Broadcast-quality FDN (Feedback Delay Network) reverb with frequency-dependent decay. Based on Fons Adriaensen's Zita reverb. The most versatile and controllable algorithm.

Common Reverb Parameters:

Parameter	Range	Default	Description
Decay	0–100%	50%	Reverb tail length
Damping	0–100%	50%	High-frequency absorption — higher values darken the reverb
Pre-Delay	0–100 ms	20 ms	Gap before reverb begins — separates dry from wet
Size	0–100%	50%	Virtual room size
Mix	0–100%	30%	Dry/wet blend

NOTE: Reverb supports spillover (Effect Trails) — when enabled in Settings, the reverb tail decays naturally even when you bypass the effect or switch presets.

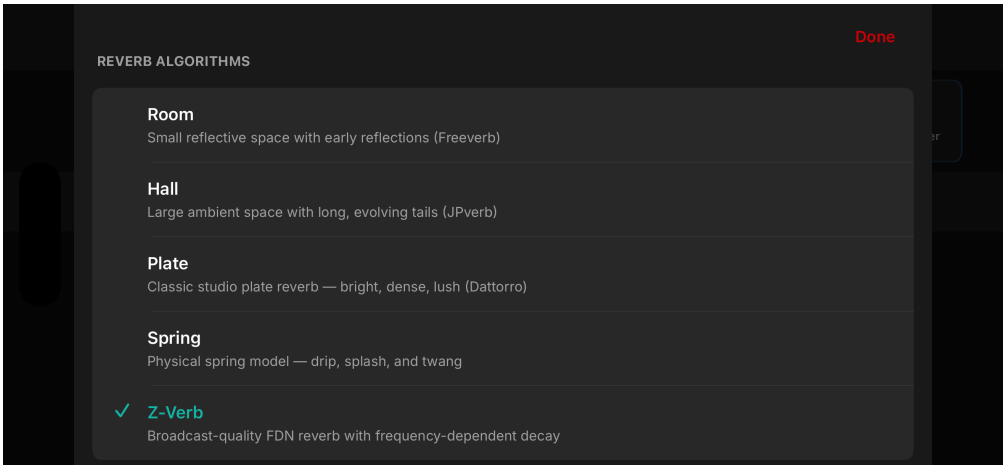

Figure 42: Reverb algorithm picker

5.8 Delay

Three delay algorithms cover the full range from pristine digital to warm analog.

5.8.1 Digital

Clean digital delay with pristine repeats and optional ping-pong stereo. Crystal-clear echoes.

5.8.2 Tape Echo

Warm, saturated repeats with wow and flutter modulation and degrading repeats. Each repeat loses high-frequency content, simulating magnetic tape saturation.

5.8.3 BBD (Bucket Brigade Device)

Analog bucket-brigade delay with bandwidth limiting. Lo-fi, band-limited character reminiscent of vintage analog delay pedals.

Common Delay Parameters:

Parameter	Range	Default	Description
Time	10–2000 ms	375 ms	Delay time between repeats
Feedback	0–100%	30%	How many repeats (higher = more echoes)
Mix	0–100%	30%	Dry/wet blend
Tone	0–100%	50%	Repeat brightness (Tape and BBD only)
Mod Rate	0–5 Hz	0.4 Hz	Wow/flutter speed (Tape and BBD only)
Mod Depth	0–100%	15%	Wow/flutter amount (Tape and BBD only)

NOTE: Delay also supports spillover — echoes trail off naturally when bypassed.

5.8.4 Tempo Sync

Delay, tremolo, and chorus support tempo sync. When enabled, the delay time locks to subdivisions of the global BPM:

Subdivision	At 120 BPM	Note Value
1/4	500 ms	Quarter note
1/8	250 ms	Eighth note
1/8 dotted	375 ms	Dotted eighth
1/16	125 ms	Sixteenth note
1/4 triplet	333 ms	Quarter triplet
1/8 triplet	167 ms	Eighth triplet

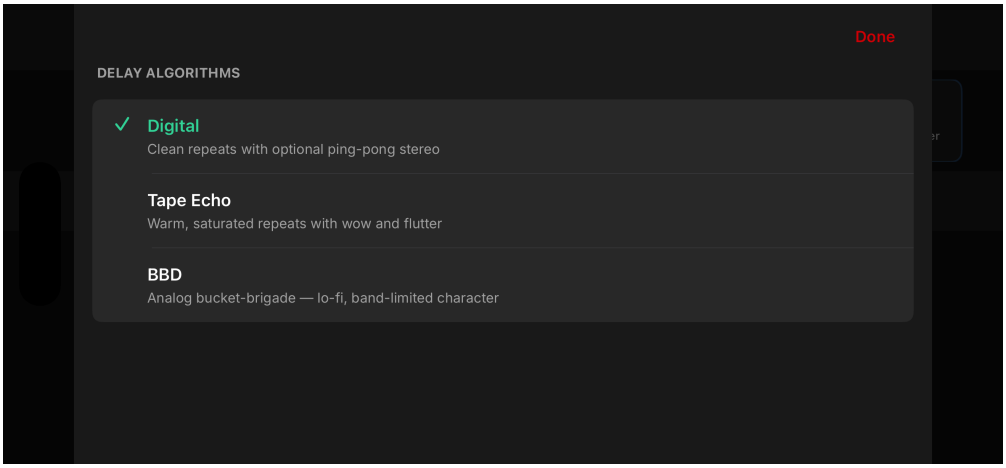

Figure 43: Delay algorithm picker

5.9 IR Loader

Loads impulse response (IR) files for cabinet simulation. IRs are short audio recordings that capture the frequency response of a guitar cabinet, microphone, and room.

Supports single and dual cab modes:

- **Single Cab** — Loads one IR file for straightforward cabinet simulation
- **Dual Cab** — Loads two IR files and blends them for complex tones

Supported file formats: `.wav`, `.aif`, `.aiff`

Ampz does not ship with bundled IR files. Import your own via the Files app share sheet, “Open In” menu, or download them from the Tone3000 tab.

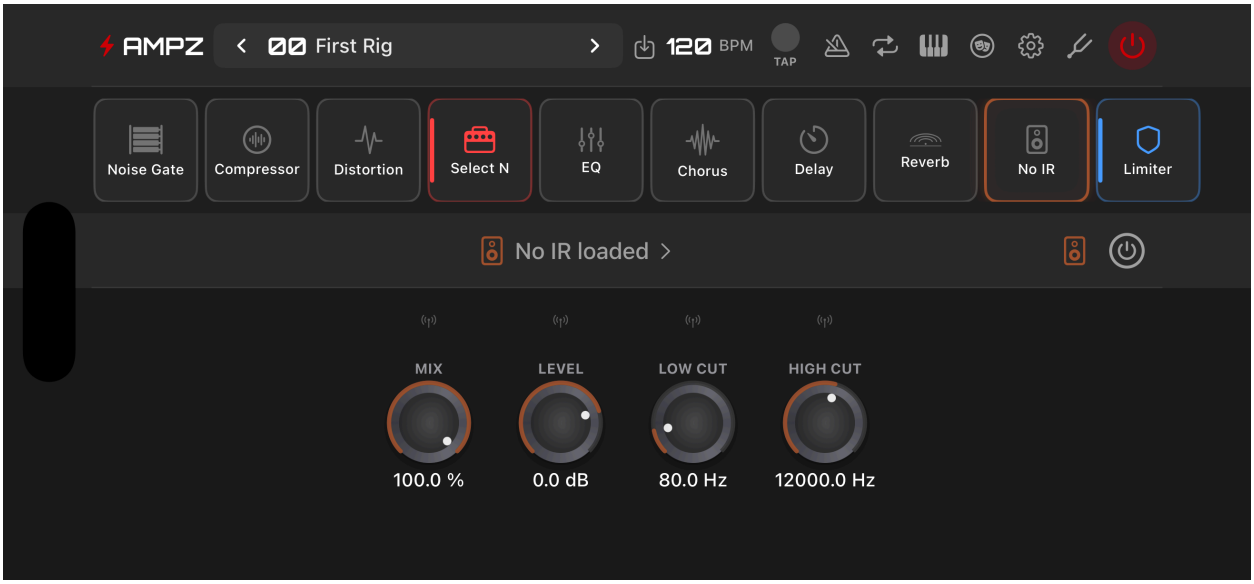

Figure 44: IR Loader effect slot

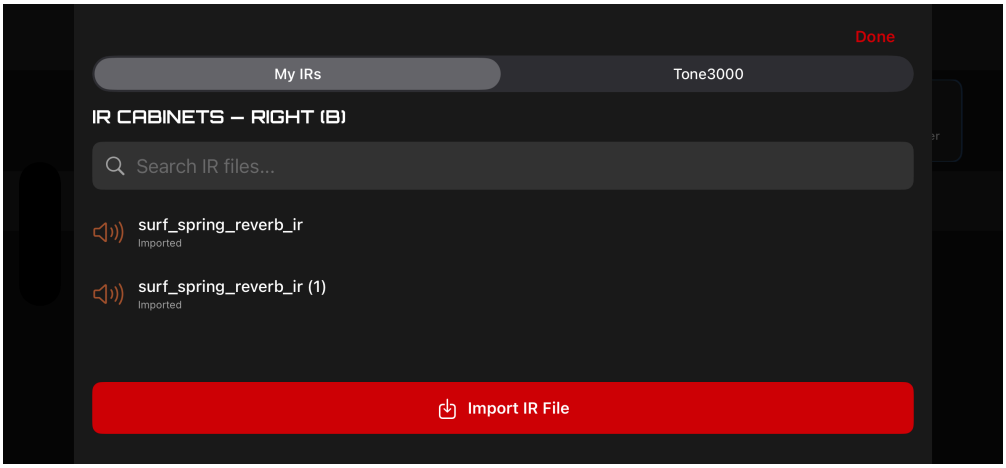

Figure 45: My IR files list

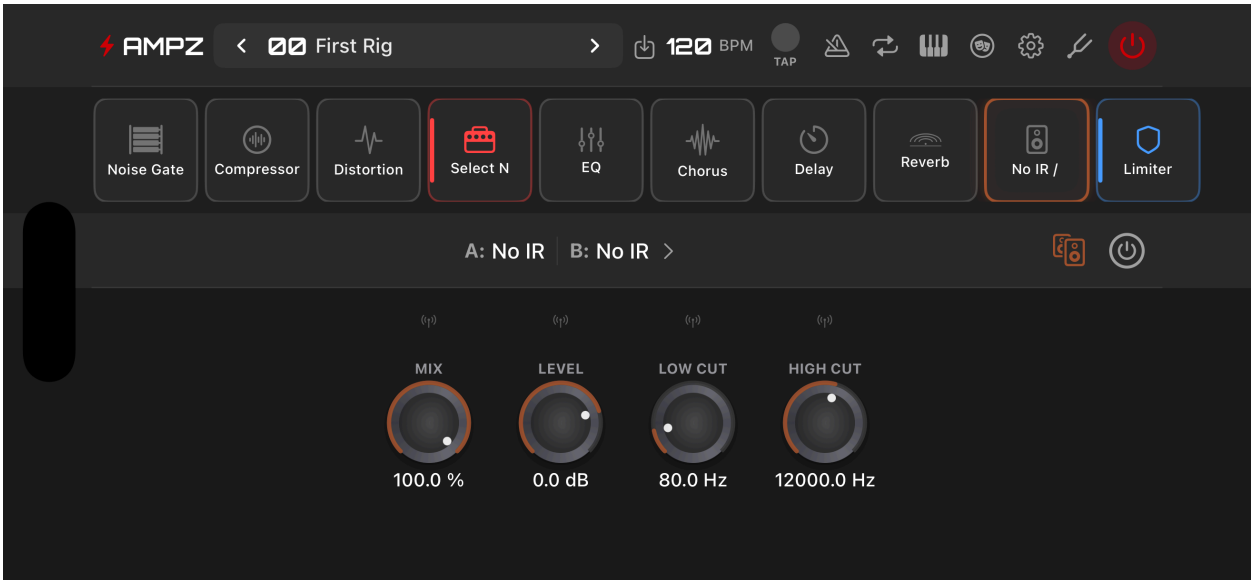

Figure 46: Dual IR toggle button

Figure 47: Dual IR view — blending two cabinet IRs

5.9.1 Downloading IRs from Tone3000

You can also download IR files from the Tone3000 platform. Navigate to the Tone3000 tab in the IR browser and search for cabinet impulse responses. The IR browser uses the same rich-card layout as the NAM browser — cover thumbnails, contributor handle, IR count per pack, downloads, and favorites — plus a sort menu for Trending / Newest / Oldest / Most Downloaded. Tap a pack to open its detail view and pick which IRs to download.

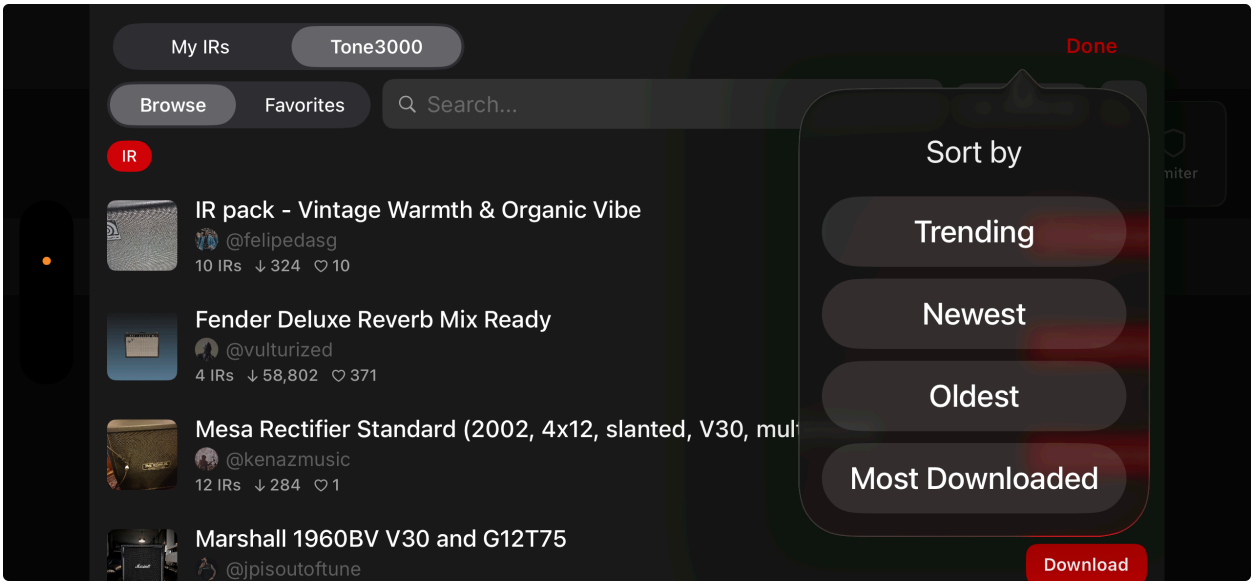

Figure 48: Tone3000 IR browser — rich cards and sort menu, mirroring the NAM browser

6 Preset System

6.1 100-Slot Architecture

Ampz provides 100 preset slots numbered 00 through 99:

- **Slots 00–09** — Factory range (Slot 00 ships with the “First Rig” starter preset; all are user-overwritable)
- **Slots 10–99** — User presets (empty by default)

Each preset stores the complete state of your rig:

- All 16 effect slot configurations (effect type, algorithm, parameters)
- Slot enable/bypass states
- The signal path: split and merge points and the Mix block settings
- All eight scenes and which one is active
- Stomp assignments that override the global map for this preset
- NAM model file references (one per amp slot)
- IR file references
- Chain-level settings (enabled, bypassed)

WARNING: Presets saved on this version need this version. A preset that uses slots 11 to 16, a split path, scenes or stomps cannot be read by an older build of Ampz, so keep every device you share rigs with up to date.

6.2 Preset Browser

Tap the preset name/number display in the toolbar to open the Preset Browser overlay. The browser provides:

- **Search bar** — Filter presets by name
- **Category filter pills** — Filter by Factory, User, or Custom categories
- **4-column grid** — Visual preset cards showing number, name, and category color
- Tap a preset card to load it immediately

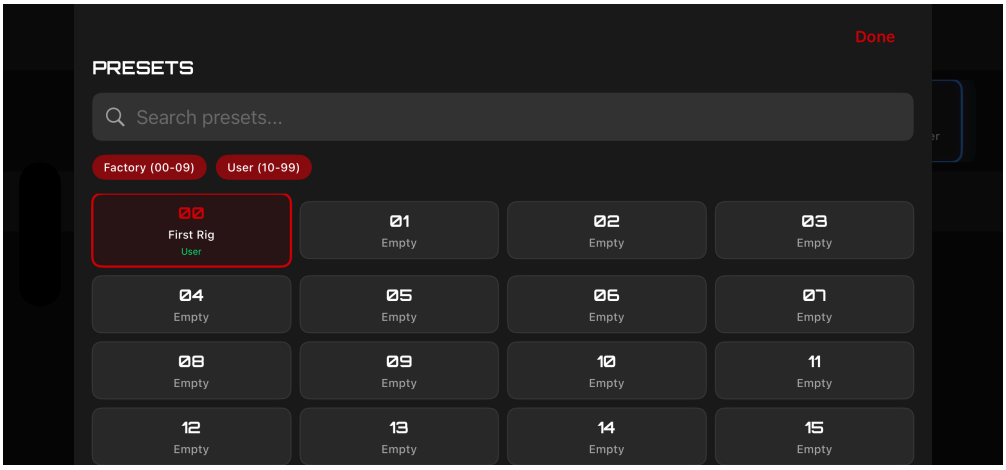

Figure 49: Preset Browser with category filter pills

6.3 Saving Presets

1. Configure your desired effects chain and parameters
2. Tap the save icon ( ) in the toolbar
3. Enter a name for your preset. When you scroll to a slot that already holds a preset, the name field auto-fills with that preset's existing name — useful for quickly overwriting in place with the same name.
4. Select a destination slot using the wheel picker
   - Empty slots show "Empty"
   - Occupied user slots show "(Overwrite)" in orange
5. Tap **Save**

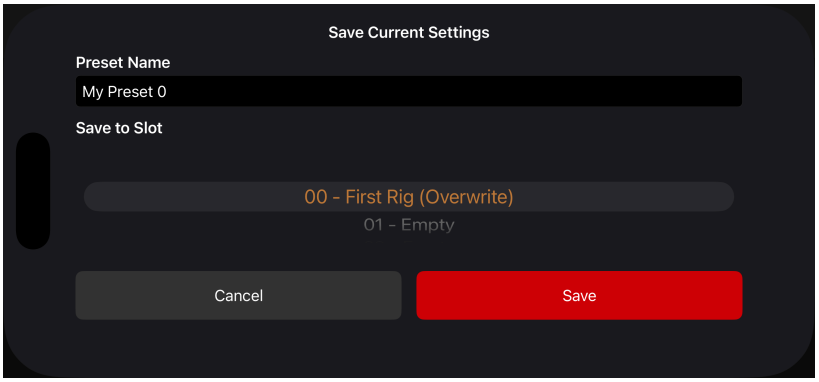

Figure 50: Saving a preset — slot picker with overwrite warning

WARNING: Factory presets in the initial slot (00) can be overwritten since "First Rig" is a user-editable starter preset. Always save to a higher slot number if you want to preserve it.

6.4 What Gets Saved

Data	Details
Effect chain	All 16 slots — type, algorithm, parameters, enable state
Signal path	Split and merge points, Mix levels, pans, crossfade, delay trim, polarity
Scenes	Eight snapshots of bypass states, parameters and Mix settings, plus the active scene
Stomps	Per-preset stomp assignments (the global default map lives outside the preset)
NAM model	File path and display name, per amp slot
IR files	File paths, names, sizes, checksums
Chain state	Global enable/bypass
Metadata	Name, created/modified dates, description

6.5 Sharing Rigs

A preset on its own is not enough to reproduce your sound somewhere else. It refers to your NAM model and cabinet IR **by name**, and those names only mean something on the device that already has the files. Send just a preset and it arrives with the amp and cab slots empty.

So Ampz shares a **rig** instead: a single `.ampzrig` file containing the preset **and** the NAM models and IRs it uses. The person on the other end opens it and plays.

6.5.1 Sharing a Rig

1. Open the preset browser
2. Press and hold the preset you want to share
3. Choose **Share Rig…**
4. Ampz tells you what the file will contain, for example “1 amp model and 1 cabinet IR”
5. Tap **Share** and pick AirDrop, Messages, Mail, or Save to Files

NOTE: The confirmation step is there on purpose. Sharing a rig copies your amp captures and cabinet IRs into a file you are about to hand to someone else, so Ampz always tells you exactly what is travelling with it before anything leaves your device.

6.5.2 Opening a Shared Rig

Either tap **Import Rig** at the top of the preset browser and choose the file, or open the `.ampzrig` straight from Files, AirDrop, Mail, or Messages and choose Ampz.

The rig lands in the **first empty preset slot**, so importing never overwrites a rig you already have. Its amp models and IRs are added to your library at the same time, exactly as if you had imported them yourself.

6.5.3 When Files Are Missing

If a rig is shared from a device that no longer has one of the referenced files, Ampz says which ones did not travel rather than loading a half-empty chain in silence. The rest of the rig still loads, so you can drop in your own amp model or cab and carry on.

TIP: A rig is also the easiest way to move a sound between your own devices. Share it to yourself by AirDrop, or save it to iCloud Drive and open it on the other device. Because Ampz stores its presets where the plugin can see them, an imported rig shows up in the AUv3 as well.

6.5.4 What a Rig Contains

Included	Details
Preset	The full 16-slot chain, signal path, scenes, stomps, parameters, and metadata
NAM models	Every amp model the preset uses, embedded in the file
Cabinet IRs	Every impulse response the preset uses, embedded in the file

WARNING: Amp captures and IRs are somebody's work, and licence terms vary from one creator to the next. A rig you share carries copies of those files. Before passing on models you downloaded rather than captured yourself, check what the creator allows.

7 Scenes

7.1 Overview

A scene is a snapshot of one rig's settings. Every preset has eight scenes, and switching between them is gapless: no pedal is rebuilt, no model is reloaded, reverb and delay tails keep ringing, and each bypass flip is crossfaded over a few milliseconds so nothing clicks. Use scenes for the sounds inside one song: clean verse, boosted chorus, lead with delay, all from one preset.

7.2 What a Scene Stores

- The bypass state of every pedal
- Every parameter of every pedal
- The Mix block's levels, pans, crossfade, delay trim and polarity

A scene does not store the pedal order, the effect types, or the split structure. Those belong to the preset, so every scene of a preset shares the same pedalboard. If you load a different effect into a slot, the scene entries for that slot are replaced by the new pedal's settings.

7.3 Switching and Editing

- **Main screen:** tap a capsule in the scene row under the meters. On short phones the row folds into a scene menu in the toolbar.
- **Live Performance:** the large scene pads above the stomp grid.
- **MIDI:** the **Scene 1** to **Scene 8**, **Scene Up** and **Scene Down** actions (see the MIDI chapter).
- **AUv3:** the plugin's **Scene** parameter, so a host can automate scene changes.

Anything you change while a scene is active sticks to that scene. Saving the preset saves all eight scenes and remembers which one was active. New presets start with eight identical scenes.

7.4 Renaming

Long-press a scene capsule (main screen or Live) and type a name. Names are saved with the preset.

> **NOTE:** **Dual Mono** and **Stereo A|B** write their pans and crossfade into every scene, so a stereo rig stays stereo when you switch scenes. Levels, delay trim and polarity remain per scene.

8 Looper

8.1 Overview

The Ampz looper provides two independent loops with a circular waveform display. Access it by tapping the repeat icon ( ) in the toolbar.

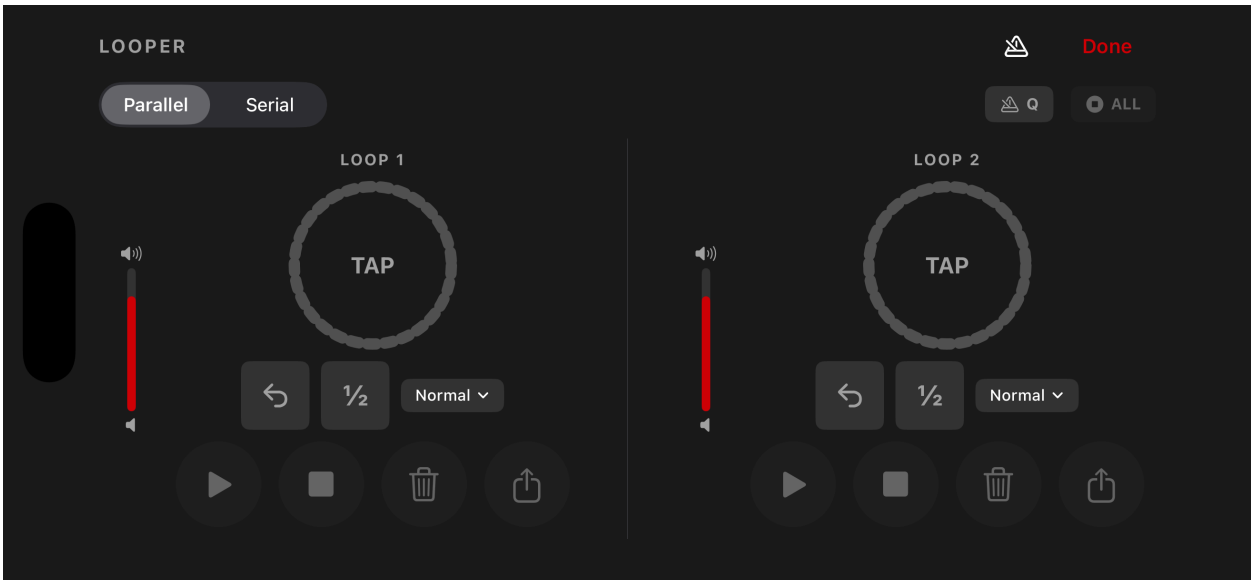

Figure 51: Looper — dual-loop interface with circular waveform displays

8.2 Dual-Loop Architecture

The looper contains two independent loop tracks (Loop 1 and Loop 2), each with its own:

- Circular waveform display showing recorded audio
- Record / Play / Stop / Clear controls
- Volume slider (vertical fader)
- Reverse and Half-Speed toggles
- Mode selector (Normal, One-Shot, Replace, Decay)

8.3 Recording Flow

1. **Tap the circle** to start recording — the border turns red and audio begins capturing
2. **Tap again** to stop recording and begin playback — the loop plays back continuously
3. **Tap while playing** to enter overdub mode — new audio layers on top of existing content
4. **Tap again** to exit overdub and resume playback

The looper records post-effects audio — your complete processed signal including NAM amp and effects.

8.4 Loop Modes

Mode	Behavior
Normal	Standard record → overdub → play cycle
One-Shot	Records one overdub pass and automatically returns to playback
Replace	Overwrites existing audio instead of layering
Decay	Previous audio gradually fades with each overdub pass

8.5 Routing Modes

A segmented control at the top switches between two routing modes:

- **Parallel** — Both loops play simultaneously, mixed together at the output. When drums are enabled, only the first recorded loop captures drum audio to prevent doubling — the second loop records guitar only.
- **Serial** — Loop 1′s output feeds into Loop 2 (Loop 1 → Loop 2 → Output). Pending cues fire at the active loop's cycle boundary for seamless transitions.

8.6 Loop Controls

- **Reverse** ( ) — Plays the loop backward in real time
- **Half Speed** (½) — Plays at half speed and one octave lower
- **Quantize** (Q) — When enabled, record start/stop snaps to BPM boundaries. A level picker appears with three options:
  - **Beat** — Snaps to the nearest beat
  - **Half Bar** — Snaps to 2-beat boundaries (half measure in 4/4)
  - **Bar** — Snaps to 4-beat boundaries (full measure in 4/4)
- **Stop All** — Stops both loops simultaneously
- **Export** ( ) — Exports the loop as an audio file via the iOS share sheet

8.7 Volume Control

Each loop has a vertical volume slider on its left side. Adjust independently to balance the mix between Loop 1 and Loop 2.

9 Drum Loops

9.1 Overview

The built-in drum loop player provides practice accompaniment with 160 professionally recorded AAC loops across 10 genres and 6 time signatures. Loops are time-stretched in real time to match your global BPM. Access it by tapping the metronome icon ( ) in the toolbar.

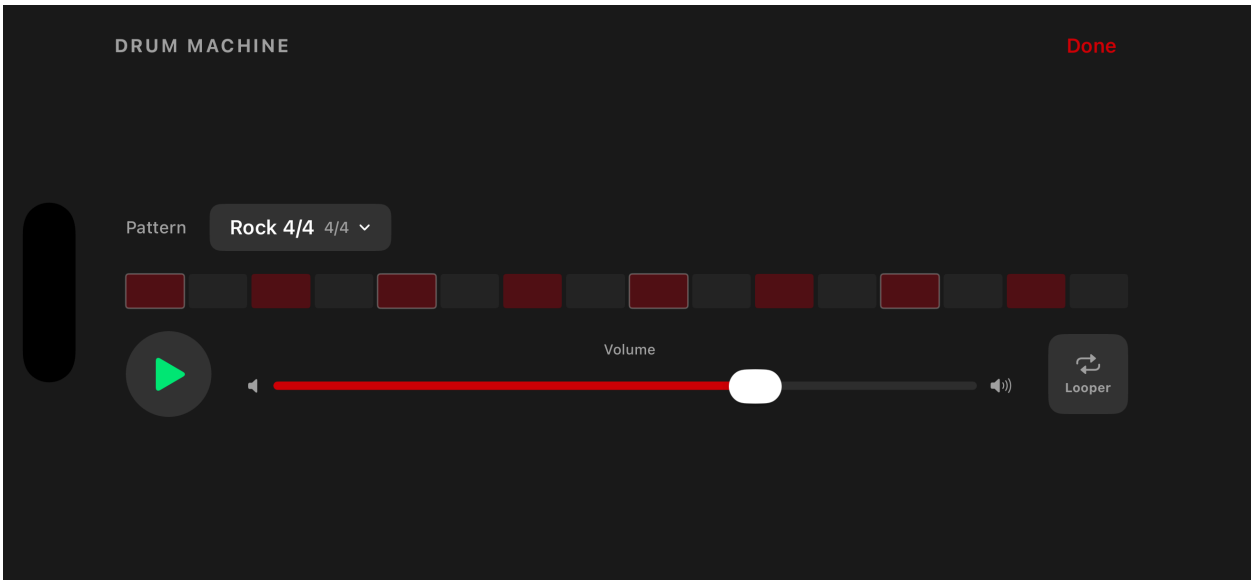

Figure 52: Drum Loops — transport controls and loop selector

9.2 Loop Browser

Tap **Select Loop** to open the loop browser. Two rows of filter chips at the top let you narrow your search:

- **Genre row** — All, Blues, Country, Funk, Jazz, Latin, Metal, Pop, R&B, Reggae, Rock
- **Time signature row** — All, 3/4, 4/4, 5/4, 6/8, 7/4, 7/8

Both filters combine — for example, selecting “Rock” and “3/4” shows only rock waltz loops. Tap a selected chip again (or tap “All”) to clear that filter.

The list is organized by genre sections. Each loop row displays the pattern name, bar count, native BPM, time signature badge (for non-4/4 loops), and kit name.

9.3 Genres & Loop Count

Genre	Loops	Styles
Blues	11	Shuffle, slow shuffle, straight, waltz, 6/8, groove, intro, fill
Country	17	Basic, two-step, waltz, 6/8, verse, intro, fill, extended
Funk	15	Basic, linear, shuffle, waltz, groove, verse, intro, fill
Jazz	17	Swing, bossa, brushes, waltz, 5/4, 7/4, groove, verse, intro, fill
Latin	13	Bossa, samba, mambo, vals (3/4), 6/8, groove, verse, intro, fill
Metal	18	Basic, blast, breakdown, double, gallop, halftime, stomp, thrash, 6/8, 7/8
Pop	15	Standard, bounce, EDM, halftime, syncopated, tropical, waltz, 6/8
R&B	8	Basic, bounce, trap, waltz, groove, verse, fill
Reggae	6	One drop, steppers, waltz, intro, fill
Rock	30	Basic, anthem, breakdown, driving, double, halftime, hard, punk, shuffle, stadium, waltz, 5/4, 6/8

Many loops include alternate variants (labeled 1, 2, etc.) for variety within the same pattern style.

9.4 Transport Controls

- **Play/Stop button** — Large circular button to start or stop the loop. Green play icon when stopped, red stop icon when playing.
- **Volume slider** — Horizontal slider to adjust loop volume relative to guitar signal
- **Looper toggle** — Include drum audio in looper recordings when enabled

9.5 Integration

- **BPM Sync** — Loops are time-stretched to match the global BPM, which can be set via Tap Tempo. The native BPM is shown next to the loop selector for reference.
- **Looper Integration** — Toggle the “Looper” button to include drum audio in looper recordings. When enabled, the drum mix is captured alongside your guitar signal.

10 Tuner

10.1 Overview

The built-in chromatic tuner provides accurate pitch detection for tuning your guitar. Access it by tapping the tuning fork icon ( ) in the toolbar.

10.2 Display

- **Note name** — Large display showing the detected note (C, C#, D, etc.) with octave number
- **Frequency readout** — Shows the exact detected frequency in Hz
- **Cents gauge** — A horizontal bar with color zones indicating tuning accuracy:
  - **Green (center)** — In tune (within ±5 cents)
  - **Amber (inner)** — Close (within ±15 cents)
  - **Red (outer)** — Out of tune (beyond ±15 cents)
- **Cents value** — Numeric display showing exact offset (e.g., “+3 cents” or “–7 cents”)

A downward-pointing triangle indicator slides along the gauge to show your current tuning position.

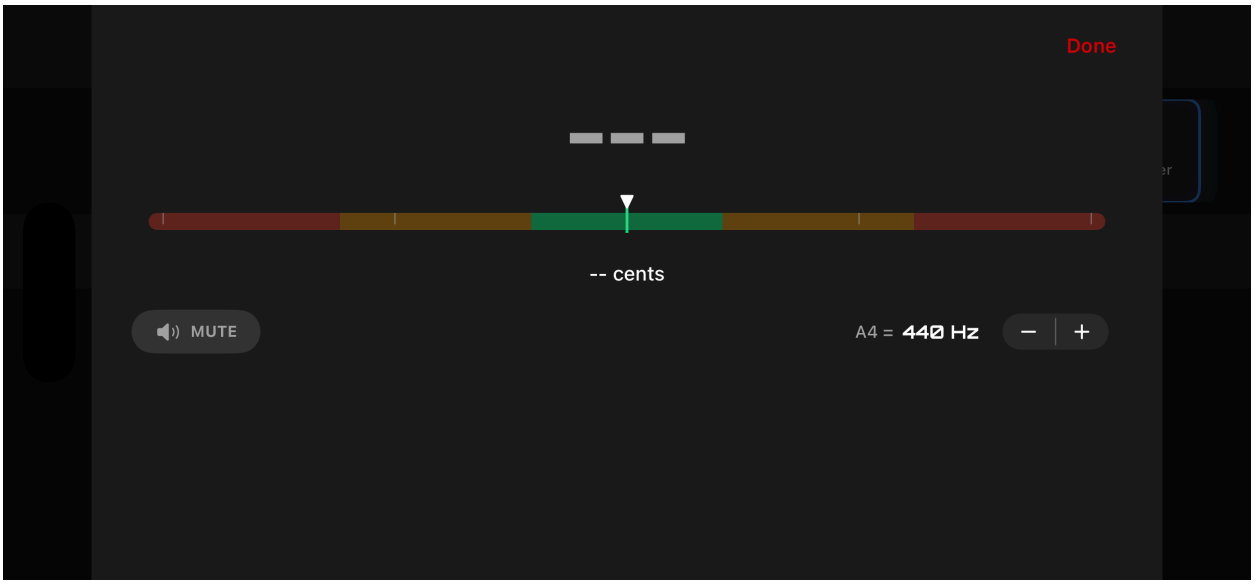

Figure 53: Tuner — note display, cents gauge, and mute control

10.3 Controls

10.3.1 Mute Output

Toggle the **MUTE** button to silence the audio output while tuning. This prevents your audience from hearing tuning noises during a performance. The button turns red when active.

When you close the tuner overlay, mute is automatically disabled.

10.3.2 Reference Pitch

Adjust the reference pitch for A4 using the stepper control:

- Range: **430 Hz to 450 Hz**
- Default: **440 Hz** (standard concert pitch)
- Use lower values (e.g., 432 Hz) for alternative tuning systems
- Use higher values for orchestral tuning contexts

11 Live Performance Mode

11.1 Overview

Live Performance mode transforms the Ampz interface into a stage-optimized view designed for foot switching and low-light visibility. Access it by tapping the theater masks icon ( ) in the toolbar.

11.2 Layout

11.2.1 Top Bar

- **"LIVE" indicator** — Amber text confirming you're in Live mode
- **Audio LED** — Green dot when audio engine is running, red when stopped
- **NAM badge** — Green "NAM" label when a neural amp model is loaded
- **Tuner button** — Quick access to the tuner without leaving Live mode
- **Done button** — Returns to the normal interface

Figure 54: Live Performance Mode — preset readout, scene pads, the PATH pad and the ten stomps

11.2.2 Preset Display

A large, high-contrast preset readout with:

- Oversized preset number (44pt font)
- Preset name with glow effect for stage visibility
- Left/right chevrons for preset navigation (tap targets sized for foot switches)

11.2.3 BPM Display

Current tempo shown in large format below the preset name.

11.2.4 Scene Pads and the PATH Pad

A row of eight large scene pads sits above the stomp grid. Tap one to switch scenes (gapless), long-press to rename. When the rig is split, an amber **PATH** pad appears at the right end of the row showing **A**, **A+B** or **B**. Tap it to jump the Mix crossfade between the lanes: a lead lane and a rhythm lane under one foot.

11.2.5 2×5 Stomp Grid

The bottom half is a grid of ten **stomps**. A stomp is a virtual footswitch that belongs to Ampz, not to a slot, so the same switch can mean “the drive pedal” in every preset even when that pedal sits in a different slot.

- **Tap a cell** to toggle its stomp
- **Long-press a cell** (or tap an unassigned one) to open the assign sheet
- A lit cell with a glowing LED is on; a dim cell is off
- A lit face with a dark LED is **partly on**: the stomp drives several pedals and they disagree
- A **×2** style tag means the stomp drives that many pedals; a lane letter means it is a path stomp
- A cell showing only its number has nothing to drive in this preset: its slot is empty, or the stomp is unassigned. Tapping an unassigned cell opens the assign sheet; a cell whose slot is simply empty ignores taps

By default stomp 1 drives slot 1, stomp 2 drives slot 2, and so on through slot 10. That default is global, and each preset can override any stomp.

11.2.6 Assigning a Stomp

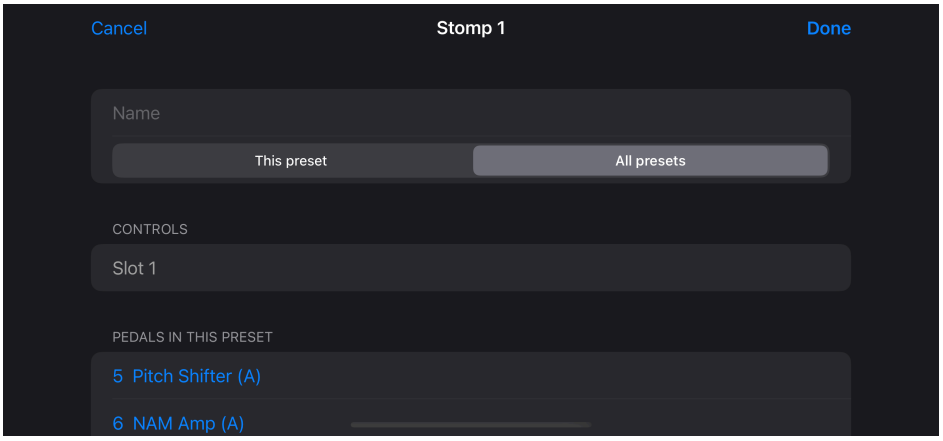

Figure 55: Stomp assign sheet — name, scope, and the three kinds of target

- **Name** — what the cell shows; leave it blank to show the pedal’s name
- **Applies to** — **This preset** overrides the stomp for the current preset only; **All presets** changes the global default
- **Pedals in this preset** — pick one pedal by its slot

- **Pedal kinds** — pick an effect kind (every Delay, every NAM Amp) with a lane filter (**Any**, **A**, **B**, **Before split**, **After merge**) and whether the stomp drives **All** matching pedals or only the **First**
- **Path** — the stomp becomes a path switch: **Path A**, **Path B**, or **Path A/B toggle**
- **Unassigned** clears the stomp

A stomp whose target matches nothing in the current preset shows as empty and ignores taps, so a mis-step on stage never opens a sheet. Stomps can be driven by MIDI through the **Stomp Mappings** tier.

11.3 Effect Abbreviations

Abbr	Effect	Abbr	Effect
NG	Noise Gate	WAH	Wah
CMP	Compressor	EQ	EQ
DST	Distortion	RVB	Reverb
AMP	NAM Amplifier	DLY	Delay
CHR	Chorus	IR	IR Loader
FLG	Flanger	LIM	Limiter
PZR	Phaser	PS	Poly Shifter
TRM	Tremolo	VIB	Vibrato

11.4 Screen Lock Prevention

Live Performance mode automatically disables the iOS idle timer, preventing the screen from dimming or locking during your performance. The idle timer is restored when you exit Live mode.

12 MIDI

12.1 Overview

Ampz supports comprehensive MIDI control for integrating with hardware MIDI controllers, foot switches, and other MIDI devices. Open Settings ( ) and tap **MIDI Settings** ( ).

Ampz organizes MIDI mappings into five tiers, each with its own identity model. Understanding the difference makes it easier to choose the right kind of mapping for what you want to do:

Tier	Keyed by	When to use
CC Mappings	Slot + parameter	Map a knob/pedal to a specific parameter (e.g. Reverb mix, Delay feedback) on whichever effect is in that slot.
Slot Bypass Mappings	Slot index	Map a footswitch to toggle a specific slot — works across all presets, controls whatever effect happens to be in that slot.
Stomp Mappings	Stomp 1–10	Map a footswitch to one of the ten Live stomps. The stomp decides what it drives in each preset, so one CC can mean “the drive pedal” everywhere.
Global Effect Mappings	Effect + algorithm	Map a footswitch to “Distortion: Klear” — toggles every Klear in your chain across all presets, regardless of slot position.
Action Mappings	App action	Map a footswitch to global actions like Tap Tempo, Preset Up/Down, Scenes, Path switching, or Looper transport.

The Mix block is a CC target too: its MIDI Learn buttons sit on the Mix panel, and a learned CC shows under **CC Mappings** as **Mix block** with the parameter name.

12.2 Enabling MIDI

Toggle the **MIDI Enabled** switch to activate MIDI input processing. When disabled, all MIDI messages are ignored.

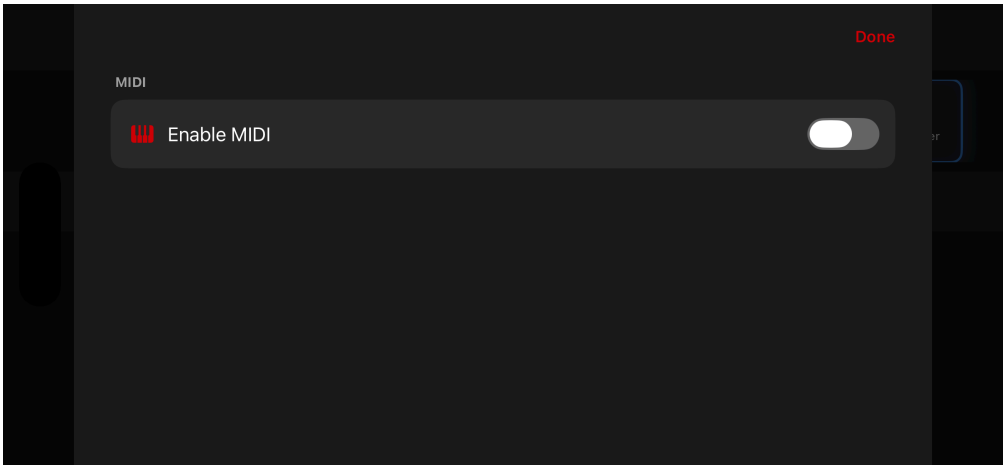

Figure 56: MIDI — enabling MIDI input

12.3 Device Discovery

When MIDI is enabled, Ampz automatically discovers connected MIDI devices:

- USB MIDI controllers connected via Lightning/USB-C adapter
- Network MIDI sessions
- Virtual MIDI ports from other apps

The Devices section lists all detected MIDI sources with their names.

12.4 Bluetooth MIDI

Tap **Bluetooth MIDI** to open the Bluetooth MIDI pairing interface. This allows you to connect wireless MIDI controllers and foot switches:

1. Enable Bluetooth on your iOS device
2. Put your MIDI controller in pairing mode
3. Tap the device name when it appears in the list
4. Once paired, the device appears in the Devices section

12.5 Active Preset

The **Active Preset** section sits at the top of the MIDI Settings screen so you can always see which effects you are mapping against. Use the chevrons to step through presets — the slot list updates in place.

Each slot row in the active preset shows:

- **Slot number and effect name** (e.g. “Slot 2 — Distortion”)
- **CC chips** listing every CC number currently mapped to that slot. Chips prefixed with a small power glyph indicate slot bypass mappings; plain `CC N` chips indicate parameter mappings. If more than three mappings exist, the row truncates with a `+N` overflow chip.
- A chevron indicating the row is tappable.

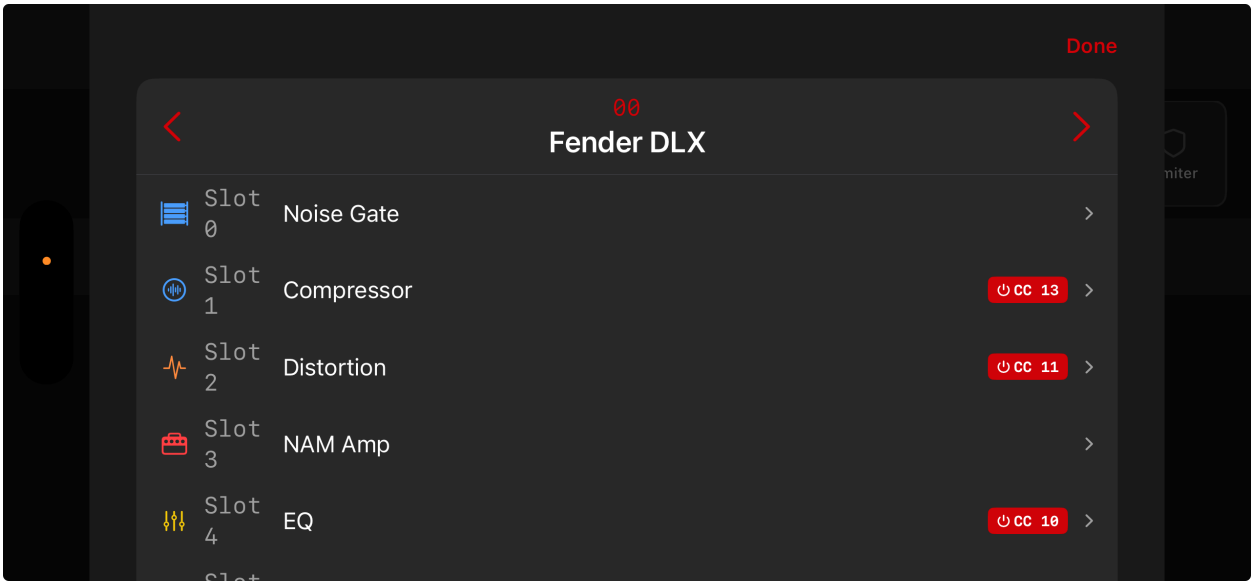

Figure 57: Active Preset section — slot rows show every CC mapped to each slot at a glance

12.5.1 Tap-to-Edit Slot Editor

Tap any slot row to open the **Slot Editor** sheet. This is the fastest way to add or change mappings for a slot you are currently looking at — no need to scroll down and re-pick the slot from a separate list.

The editor shows two sections:

- **BYPASS** — A Latch/Toggle mode picker and either the existing bypass mapping (with trash and Relearn buttons) or a row reading “Not mapped” with a Learn button.
- **PARAMETERS** — Every parameter on the slot’s effect, one row each. Each row shows the parameter name, its range, the currently mapped CC (if any), and a small antenna icon for MIDI Learn.

A footer caption explains that mappings here apply to the slot position across all presets; to map by effect type instead, use Global Effect Mappings further down the screen.

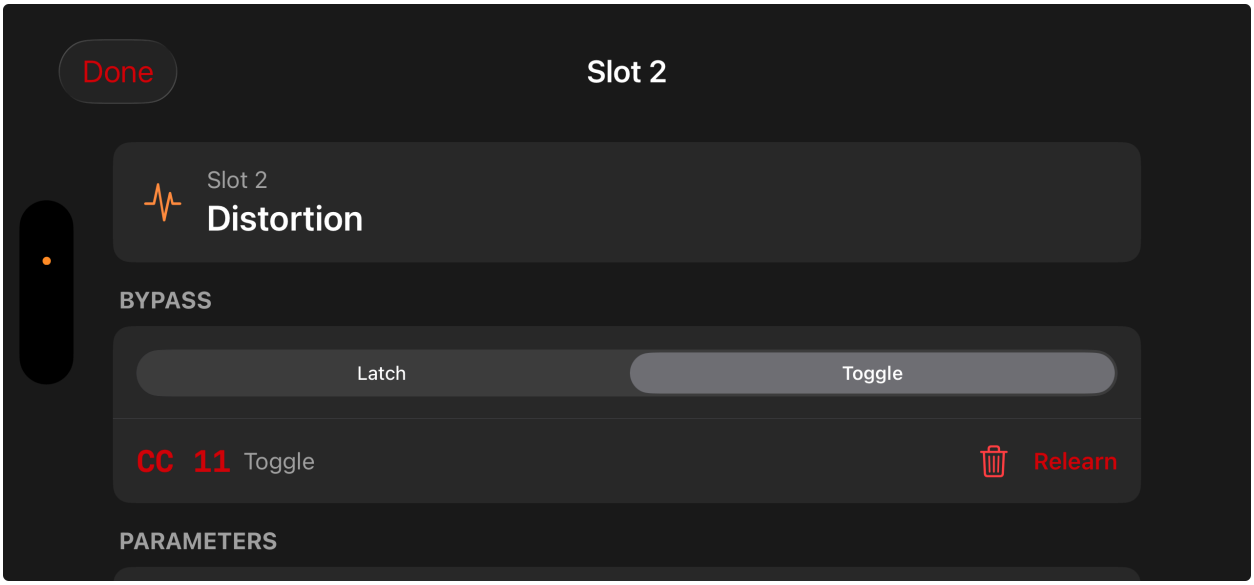

Figure 58: Slot Editor — bypass mode picker, mapped CC with Relearn, and the parameters list

12.6 Program Change

When **Program Change** is enabled, receiving a MIDI Program Change message switches to the corresponding preset number:

- Program Change 0 → Preset 00
- Program Change 1 → Preset 01
- …up to Program Change 99 → Preset 99

This allows you to switch presets from a MIDI foot controller.

12.7 CC Mappings

CC (Continuous Controller) mappings let you control effect parameters with MIDI knobs, faders, or expression pedals.

Each CC mapping connects:

- A **MIDI CC number** (0–127) to
- A **parameter** on a specific **effect slot**

Example: Map CC 1 (Mod Wheel) to the Reverb mix parameter, or CC 7 to the Delay feedback.

> **TIP:** For per-parameter mappings, opening the Slot Editor from the Active Preset section is usually faster than the dedicated Add CC Mapping flow — both end up creating the same mapping.

12.8 Slot Bypass Mappings

Slot bypass mappings let you toggle a specific **slot position** on or off with MIDI — independently of whichever effect is loaded in that slot. The mapping is global: it follows the slot across every preset.

Each row in the list reads simply “Slot N Bypass” with a Latch or Toggle subtitle. The effect name is intentionally omitted because it would change every time you switched presets.

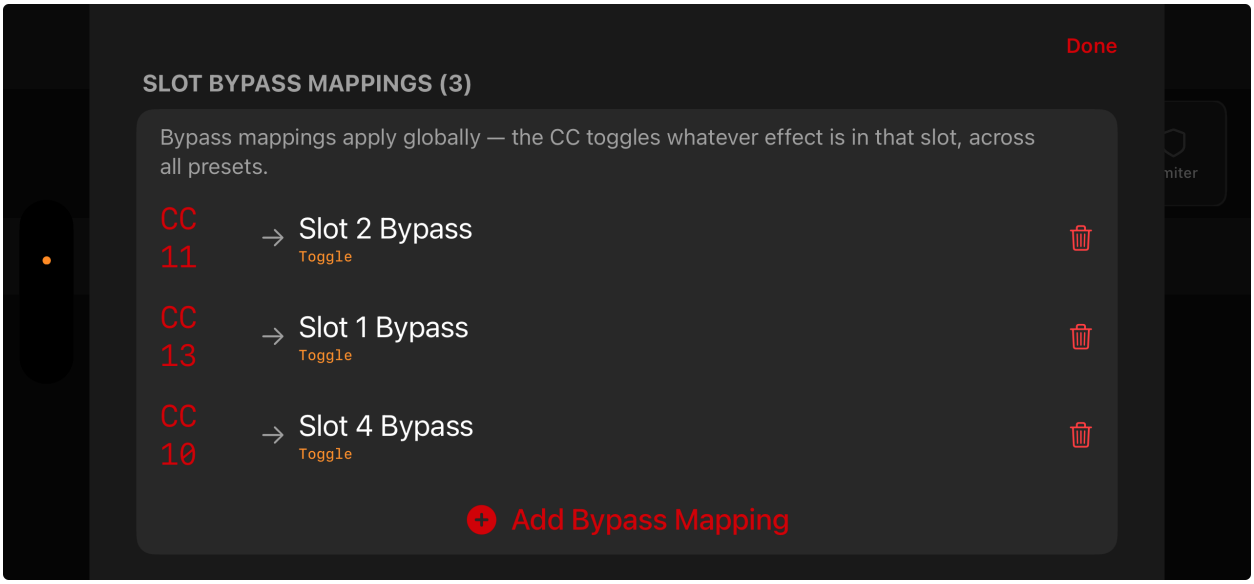

Figure 59: Slot Bypass Mappings — keyed by slot index, not by effect

12.8.1 Adding a Bypass Mapping

Tap **Add Bypass Mapping** to open the slot picker. All 16 slots are listed, even ones currently empty — because you may want to pre-assign a CC for a slot you have not populated yet in the active preset. When that slot is later filled in any preset, the mapping fires automatically.

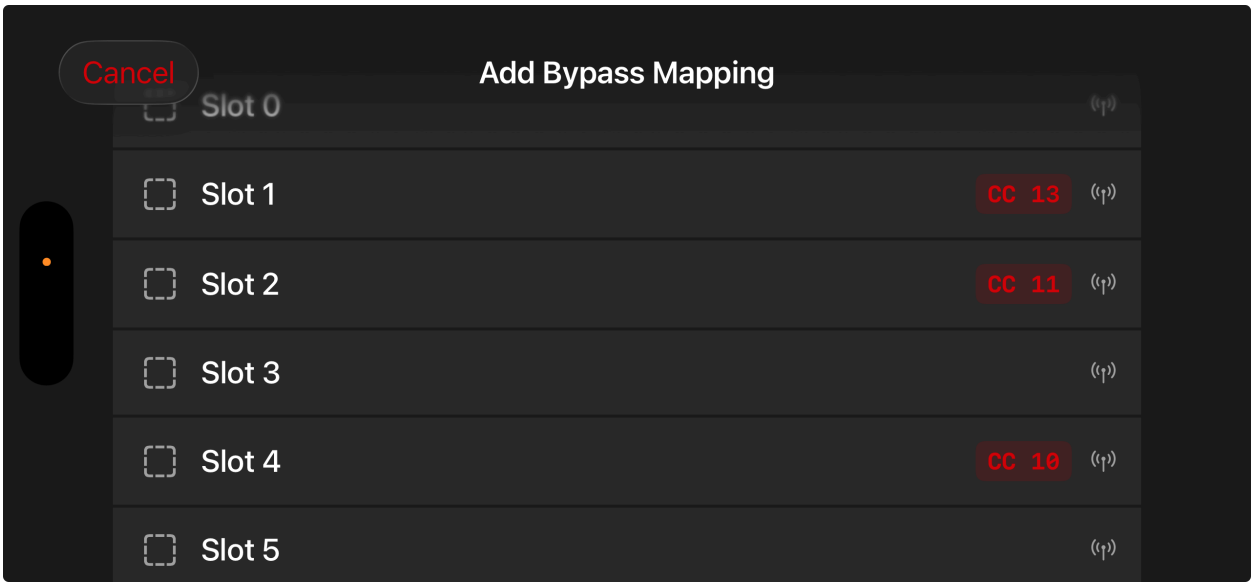

Figure 60: Bypass Slot Picker — every slot is pickable, with any existing CC chip shown

After picking a slot, choose the bypass mode:

- **Latch** — CC ≥ 64 = effect ON, CC < 64 = effect OFF. Use with continuous controllers and latching footswitches.
- **Toggle** — Each press (CC ≥ 64) flips bypass on/off. Release (CC < 64) is ignored. Use with momentary footswitches.

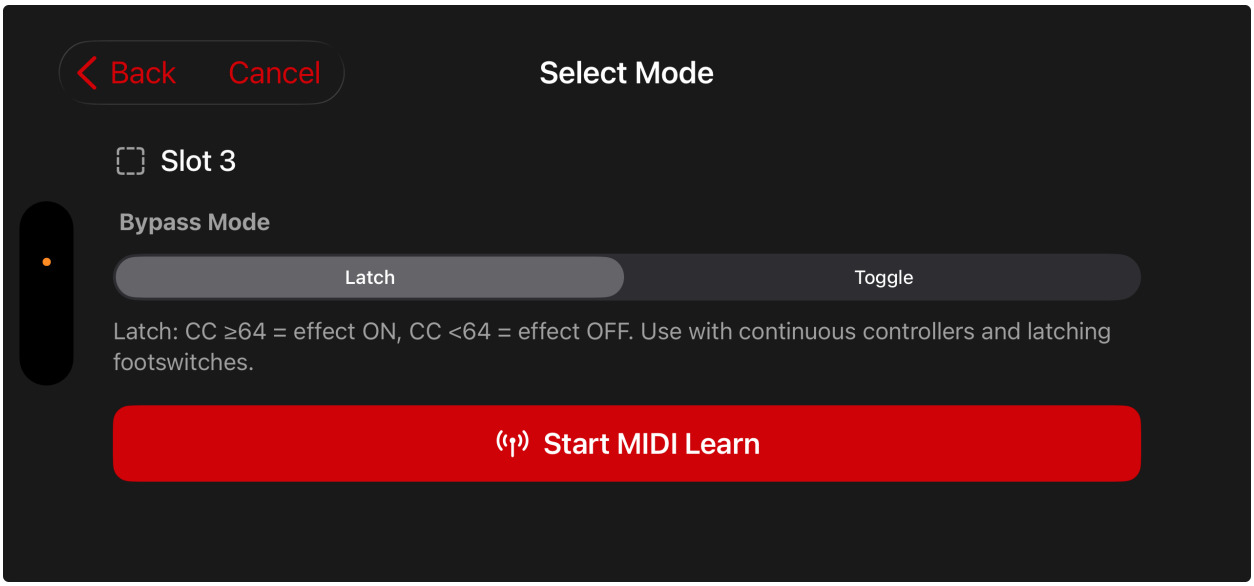

Figure 61: Mode picker — pick Latch or Toggle, then Start MIDI Learn

Tap **Start MIDI Learn** and move the desired controller. The CC number is captured automatically.

12.9 Stomp Mappings

Stomp mappings drive the ten stomps from the Live Performance grid. A CC learned here toggles the same stomp in every preset, and the stomp resolves to whatever it is assigned to there: one pedal, every pedal of a kind, or a path switch. Tap **Add Stomp Mapping**, pick the stomp, choose **Latch** or **Toggle**, then **Start MIDI Learn**.

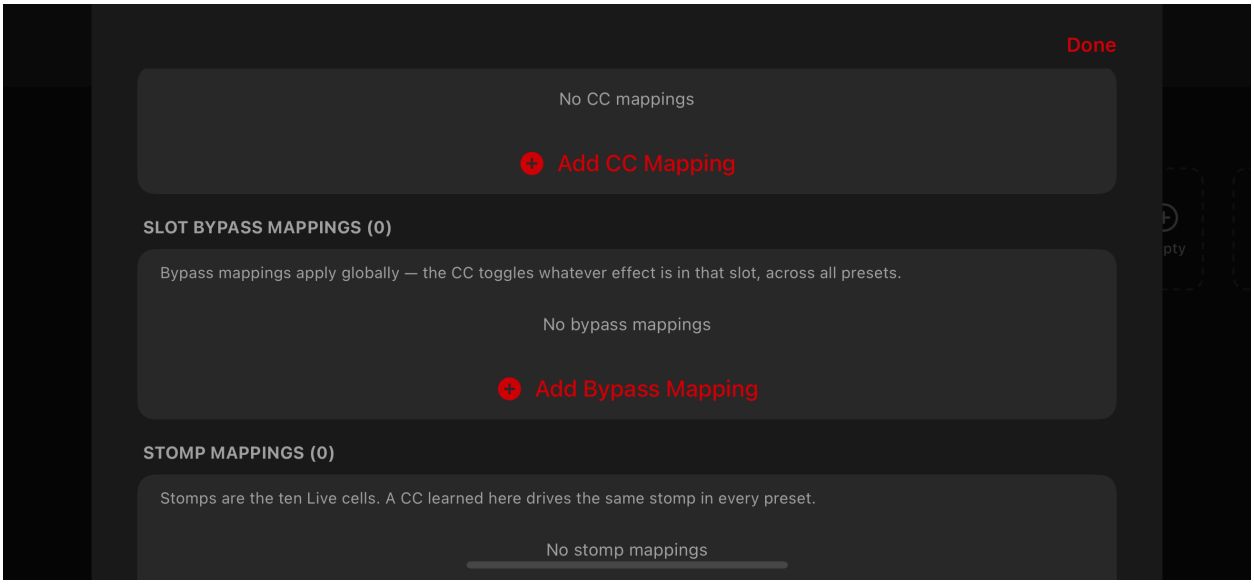

Figure 62: Stomp Mappings — the fifth tier sits between Slot Bypass and Global Effect mappings

NOTE: Precedence for one CC: a Slot Bypass mapping wins over a Stomp mapping, which wins over a Global Effect mapping.

12.10 Global Effect Mappings

Global Effect Mappings target effects by their **type and algorithm** — not by slot. A CC mapped to “Distortion: Klear” toggles every Klear instance loaded in the active chain, regardless of which slot it lives in or which preset you are on. This is what you want when you treat your pedalboard mentally as a set of pedals: the same physical button always toggles the same **effect**, even if it has moved slots between presets.

If multiple matching effects exist in a single chain (e.g. two Tape Echo delays), all of them toggle together from that one CC.

The list shows the full catalog of effects and algorithms — one row per `(effect type, algorithm)` pair. For effects without sub-algorithms (Noise Gate, Compressor, Flanger, Limiter, NAM Amp, IR Loader), a single row represents the entire effect. Each row shows where the effect appears in your saved presets (e.g. “In Preset 00, 01, 04”) so you know which presets the mapping will affect.

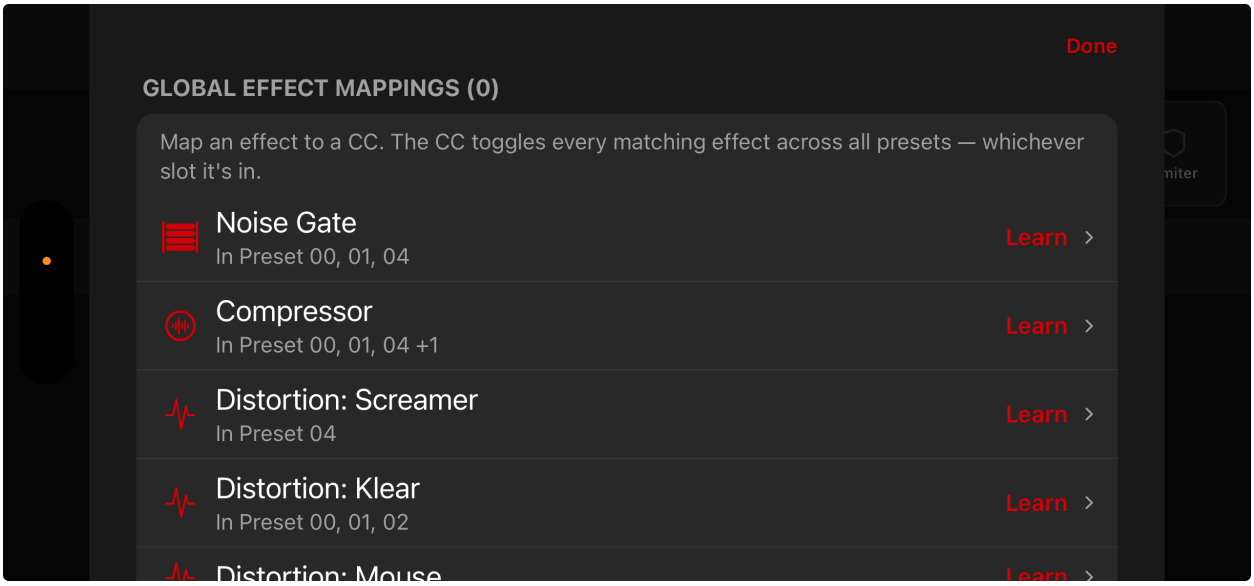

Figure 63: Global Effect Mappings — full effect catalog, each row keyed by effect + algorithm

12.10.1 Mapping an Effect

Tap any row to open the **Map Effect** sheet. The header shows the effect with the same “In Preset N” preset list, followed by the Latch/Toggle bypass mode picker and a **Start MIDI Learn** button.

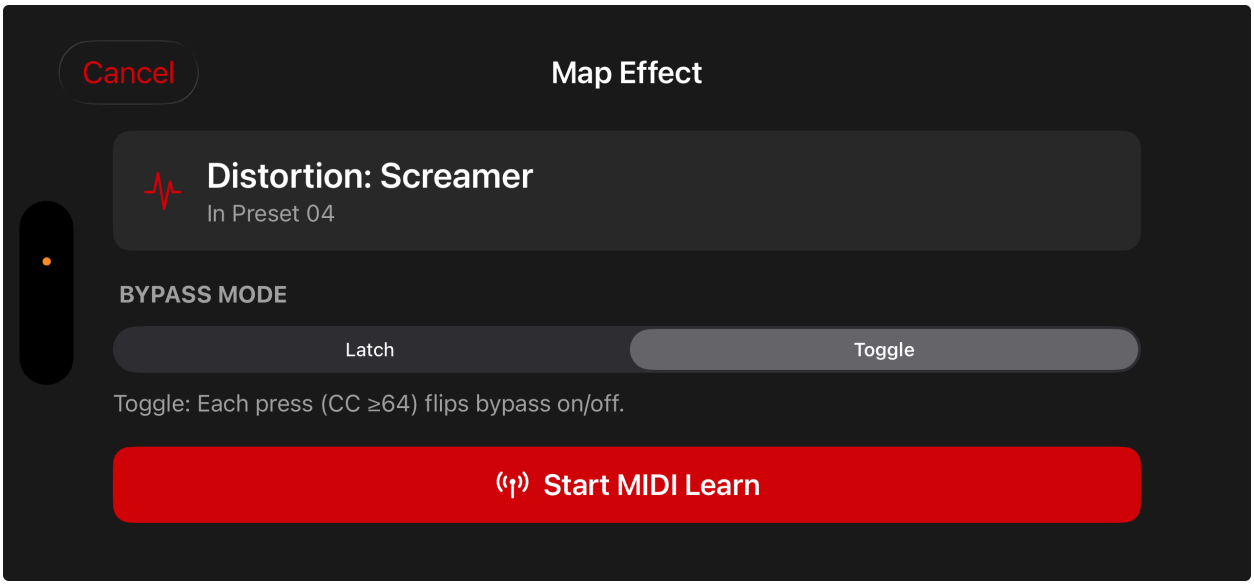

Figure 64: Map Effect — pick a bypass mode, then start MIDI Learn

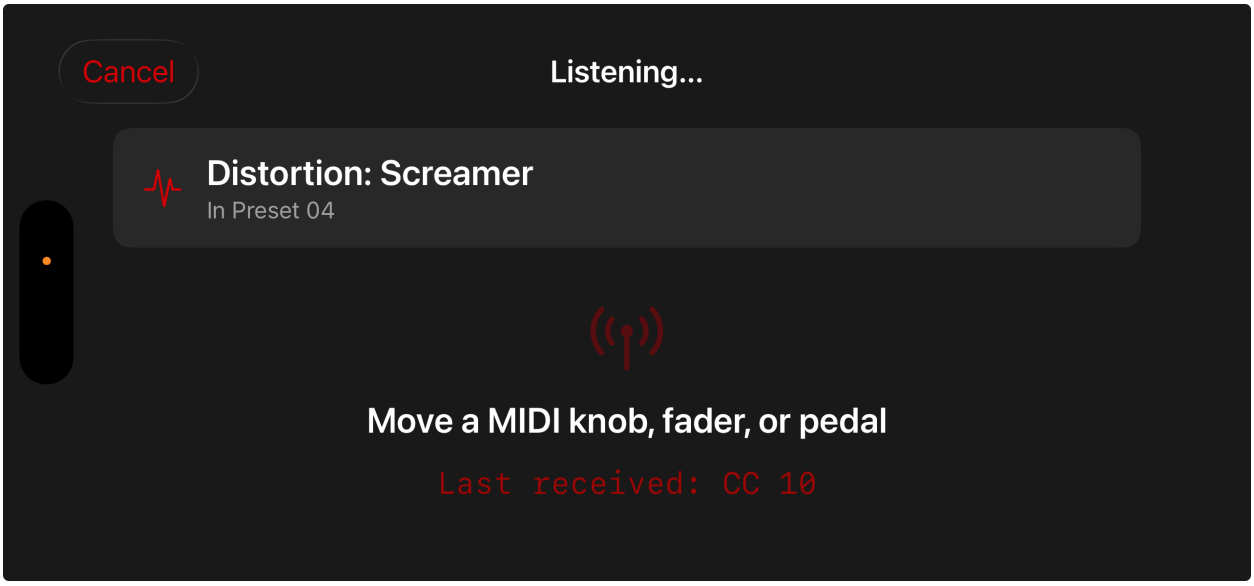

Figure 65: Listening view — move a MIDI knob, fader, or pedal to capture the CC

NOTE: Precedence: If a CC has both a Slot Bypass Mapping **and** a Global Effect Mapping bound to it, the slot mapping wins. This lets you keep a stable global mapping while overriding it for specific slots without double-toggling the same effect.

12.11 Action Mappings

Action mappings assign MIDI CC numbers to app-level actions. Actions are **momentary triggers** — they fire once each time a CC message with a value of 64 or higher is received. CC values below 64 are ignored.

Action	Description
Tap Tempo	Set BPM by repeated CC triggers
Preset Up	Load the next preset slot
Preset Down	Load the previous preset slot
Scene 1 … Scene 8	Switch to that scene of the current preset
Scene Up / Scene Down	Step through the scenes
Path A	Move the Mix crossfade fully to lane A
Path B	Move the Mix crossfade fully to lane B
Path A/B toggle	Jump between the lanes (A+B counts as A)
Loop 1 Rec/Dub	Start recording or toggle overdub on Loop 1
Loop 1 Play	Start/stop playback on Loop 1
Loop 1 Clear	Erase Loop 1 audio

Action	Description
Loop 1 Reverse	Toggle reverse playback on Loop 1
Loop 1 Half-Speed	Toggle half-speed on Loop 1
Loop 1 Stop	Stop Loop 1
Loop 1 Undo	Undo the last overdub pass on Loop 1
Loop 2 Rec/Dub	Start recording or toggle overdub on Loop 2
Loop 2 Play	Start/stop playback on Loop 2
Loop 2 Clear	Erase Loop 2 audio
Loop 2 Reverse	Toggle reverse playback on Loop 2
Loop 2 Half-Speed	Toggle half-speed on Loop 2
Loop 2 Stop	Stop Loop 2
Loop 2 Undo	Undo the last overdub pass on Loop 2
Looper Stop All	Stop both loops simultaneously

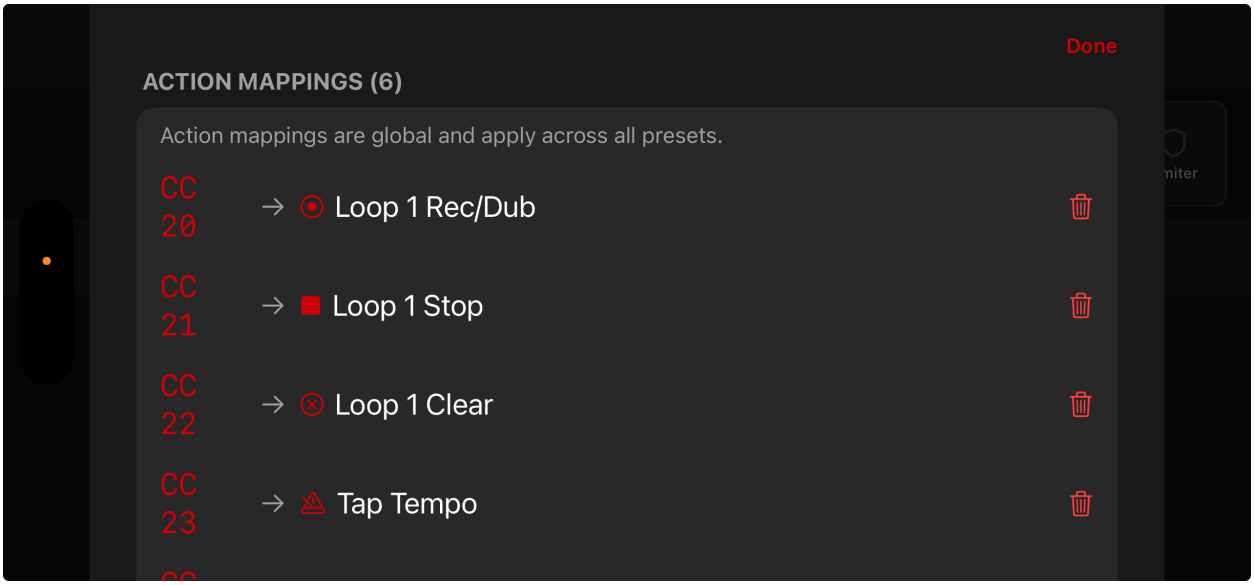

Figure 66: Action Mappings — global app actions like tap tempo and looper transport

12.11.1 Controller Configuration for Actions

Configure your MIDI foot controller buttons in **momentary** mode (sometimes called “trigger” or “pulse”) so that each press sends a single CC message with value 127. Do **not** use toggle mode (which alternates between 0 and 127 on successive presses) — toggle mode causes every other press to be ignored because the value-0 message falls below the threshold.

Mode	Behavior	Result
Momentary	Every press sends 127	Every press triggers the action
Toggle	Alternates 127 → 0 → 127 → 0	Every other press is ignored

Note: Slot Bypass, Stomp and Global Effect mappings support both momentary and toggle modes because they use separate on/off thresholds. Action mappings (looper, tap tempo, preset and scene navigation, path switching) only respond to momentary triggers.

12.12 MIDI Learn

Every mapping type uses the same learn flow:

1. Tap the **Learn** button or **Start MIDI Learn** in the relevant mapping flow
2. The screen enters a “Listening…” state with a pulsing antenna icon
3. Move the desired knob, fader, or pedal on your MIDI controller
4. Ampz captures the CC number and creates the mapping automatically

To re-learn a mapping with a different controller, tap **Relearn** on the existing mapping row (in the Slot Editor) or tap the row again from the section list.

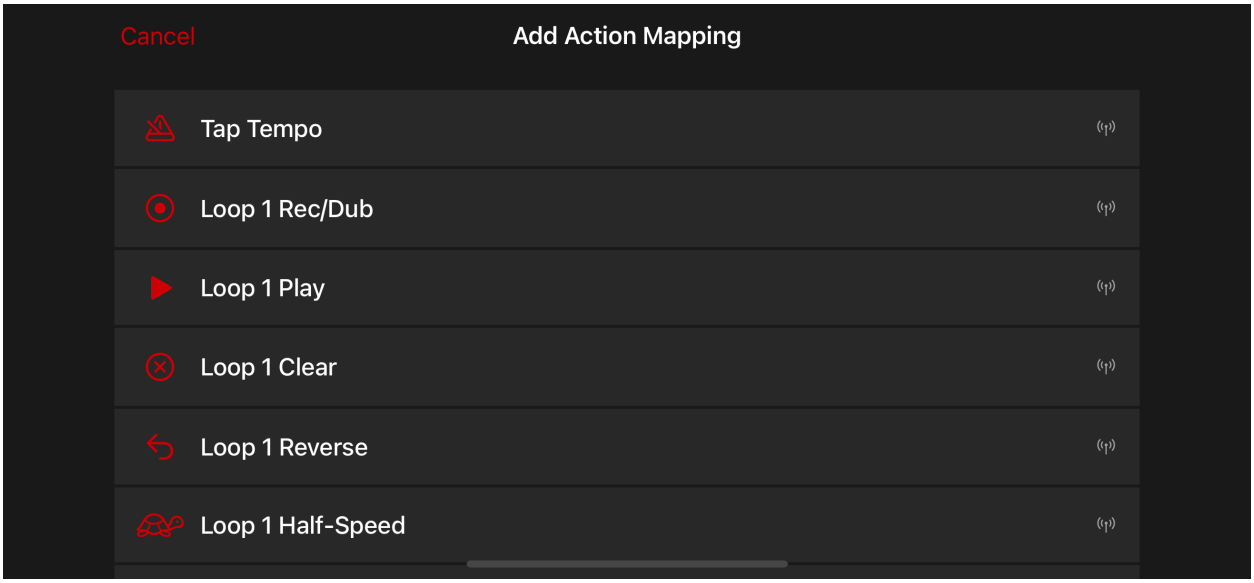

Figure 67: Add Action Mapping — scroll the list for the Scene and Path actions below the looper transport

13 Settings

Access settings by tapping the gear icon ( ) in the toolbar.

13.1 Audio Configuration

13.1.1 Input/Output Channels

When a multi-channel audio interface is connected, use the channel selector to choose which input and output channels Ampz uses. The interface name and available channels are displayed automatically.

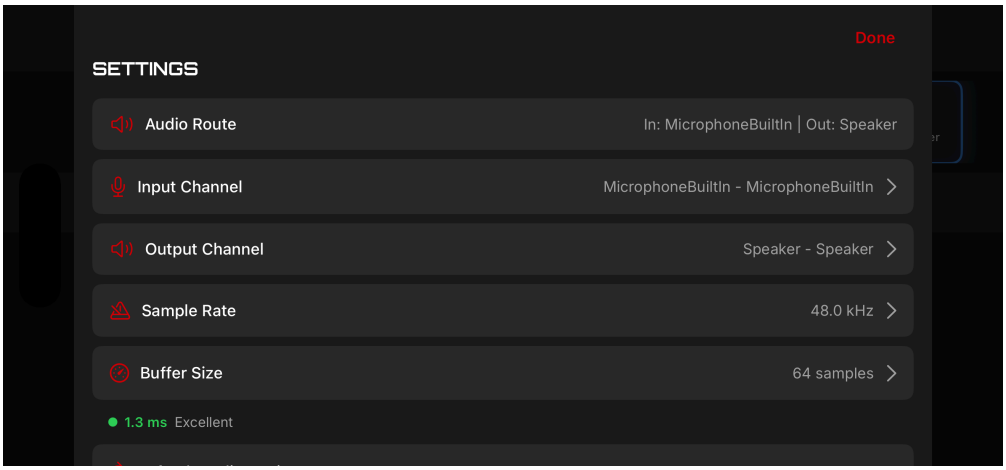

Figure 68: Settings — audio configuration

13.1.2 Sample Rate

Rate	Description
44.1 kHz	CD Quality — better device compatibility
48.0 kHz	Professional — lower latency (recommended)
96.0 kHz	Hi-Res — requires more CPU, may not be supported on all devices

13.1.3 Buffer Size

Size	Latency @ 48 kHz	Description
32 samples	0.67 ms	Ultra-low latency — requires powerful device
64 samples	1.33 ms	Professional — recommended balance
128 samples	2.67 ms	Standard — good for most devices

Size	Latency @ 48 kHz	Description
256 samples	5.33 ms	Safe — best compatibility

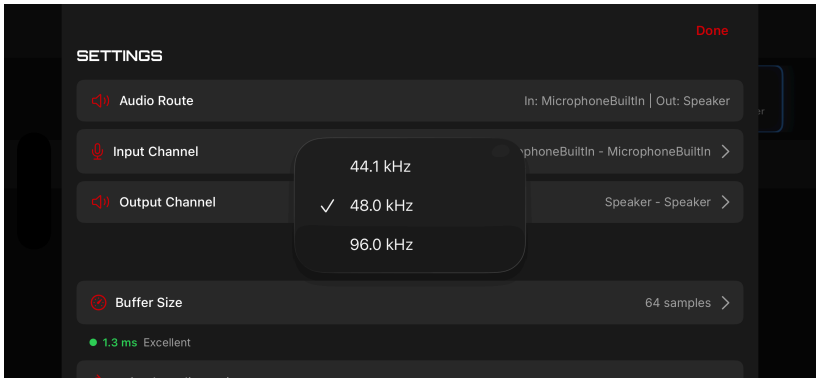

Figure 69: Settings — sample rate picker

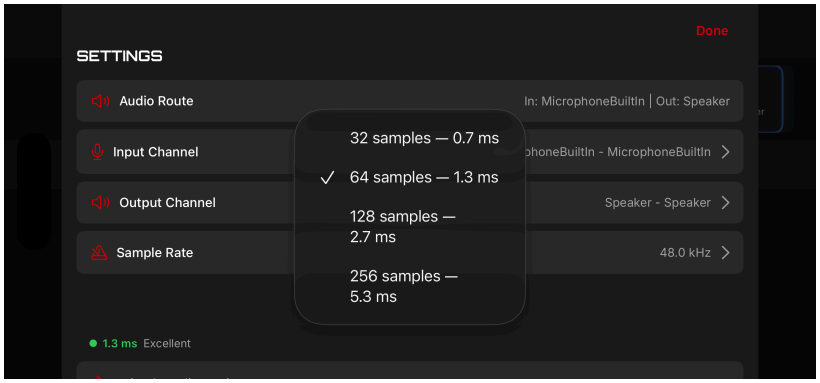

Figure 70: Settings — buffer size picker

13.1.4 Latency Display

The settings overlay shows computed one-way buffer latency based on your selected sample rate and buffer size, along with a performance rating:

- **Excellent** (green) — ≤2.5 ms
- **Good** (orange) — 2.5–5.0 ms
- **High** (red) — >5.0 ms

13.2 Background Audio

Toggle **Background Audio** to keep Ampz processing when the app is minimized or the screen is locked. Essential for live performance and practice sessions where you might glance at another app.

13.3 Effect Trails (Spillover)

Toggle **Effect Trails** to enable reverb and delay tail spillover. When enabled:

- Reverb tails decay naturally when you bypass the reverb or switch presets

- Delay echoes trail off instead of cutting abruptly
- Creates smoother transitions during live performance

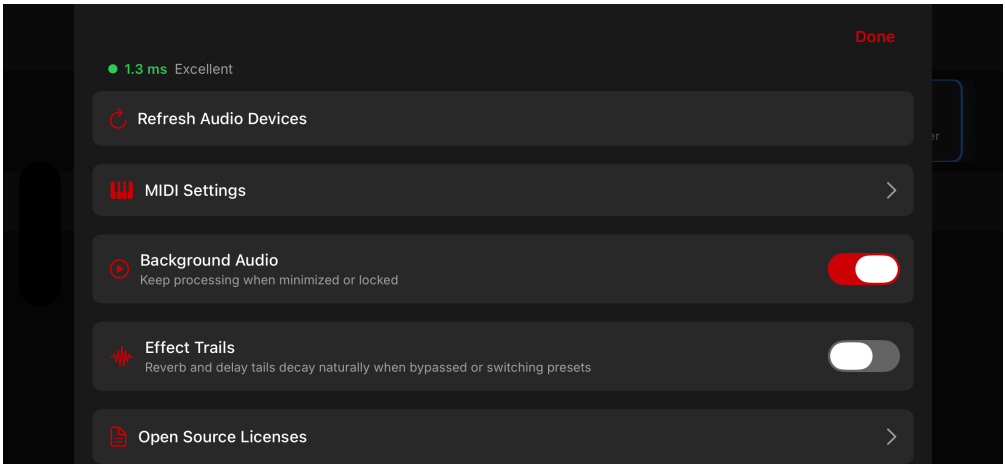

Figure 71: Settings — Background Audio, Effect Trails, and more

13.4 MIDI Settings

Tap **MIDI Settings** to navigate to the full MIDI configuration screen (see the MIDI chapter).

13.5 Open Source Licenses

Tap **Open Source Licenses** to view attribution and license information for open-source libraries used in Ampz:

- NeuralAudio (NAM processing)
- Eigen (linear algebra)
- And other components

14 Tips & Troubleshooting

14.1 Signal Chain Best Practices

A well-ordered effects chain makes a big difference in tone quality. Here's a recommended signal flow:

Slot	Effect	Why
1	Noise Gate	Remove noise before it gets amplified
2	Compressor	Even out dynamics before distortion
3	Wah / EQ	Shape the input signal
4	Distortion	Overdrive / fuzz on a clean, gated signal
5	NAM Amplifier	Core amp tone
6	Chorus / Flanger / Phaser	Modulation after amp sounds more natural
7	EQ	Post-amp tone shaping
8	Delay	Echoes after amp for cleaner repeats
9	Reverb	Space and ambience last
10	Limiter	Protect output from clipping

TIP: This is a starting point, not a rule. Experiment with different orders — putting reverb before distortion creates shoegaze textures, and compression after distortion can tighten up your sound.

For a two-amp rig, tap **IN** and choose **Middle Split**: gate, compressor and drive in front, an amp and IR in each lane, delay, reverb and limiter after the merge. **Dual Mono** puts the amps in separate speakers; a centred crossfade blends them in mono.

14.2 Latency Optimization

Low latency is critical for feeling connected to your instrument. Here's how to minimize it:

1. **Use 48 kHz / 64 samples** — The recommended balance of quality and speed
2. **Use a USB audio interface** — Dedicated interfaces provide the lowest driver latency
3. **Close other audio apps** — Background audio processing competes for CPU
4. **Use Release builds for testing** — Debug builds add overhead (relevant for developers only)
5. **Reduce effects chain length** — Each active effect adds a small amount of processing time
6. **Measure before guessing** — tap the DSP badge under the toolbar. The panel shows what each pedal costs, and the benchmark times every NAM model in your library. A full-size amp capture is the usual culprit; its Quality knob trades fidelity for headroom

14.3 Common Issues

14.3.1 No Audio Output

1. Check that audio input permission is granted (Settings → Privacy → Microphone → Ampz)
2. Verify the green LED is pulsing in the toolbar (audio engine running)
3. Check that Global Bypass (power button) is **not** active
4. If using an external interface, check the input/output channel selection in Settings
5. Try toggling the audio engine off and on (leave and re-enter the app)

14.3.2 High Latency / Crackling

1. Reduce buffer size in Settings (try 64 or 128 samples)
2. Lower the sample rate to 48 kHz if using 96 kHz
3. Close background apps that may be using audio
4. Restart the app to reset the audio session
5. On older devices, reduce the number of active effects

14.3.3 NAM Model Won't Load

1. Verify the `.nam` file is a valid NAM model (not corrupted during transfer)
2. Check that the file is in the correct format (JSON-based NAM format)
3. Try re-importing the model via the Files app
4. Some very large models may require more memory — try closing other apps

14.3.4 Preset Shows "Missing Files"

When a preset references a NAM model or IR file that has been moved or deleted:

1. Re-import the missing file
2. Or save a new preset with an updated file reference
3. Ampz will suggest alternative files with similar names if available

14.3.5 MIDI Controller Not Detected

1. Ensure MIDI is enabled in Settings
2. For USB MIDI: check the Lightning/USB-C adapter connection
3. For Bluetooth MIDI: verify Bluetooth is enabled and the controller is in pairing mode
4. Try disconnecting and reconnecting the MIDI device

15 The Ampz AUv3 Plugin

Ampz also ships as an **Audio Unit v3 (AUv3) effect plugin** (type `aufx ampz LHNE`) that is embedded inside the app. Once you have installed Ampz, the plugin is automatically available to any AUv3 host on your device — no separate download. It runs the same DSP engine as the standalone app, so your tones translate directly.

15.1 What the Plugin Includes

The plugin is a full effect insert, not a cut-down version:

1. The complete 16-slot effects chain — every effect and algorithm from the standalone app, including split lanes, the Mix block, scenes and the Signal Path templates (tap **IN** in the plugin's strip for the Chain Overview).
2. **Per-slot multi-NAM** — any slot can host its own `.nam` model, exactly as in the app.
3. Host-tempo sync for time-based effects (delay, tremolo, and other synced effects follow the host's BPM).
4. Host-automatable parameters (see *Host Automation* below).

NOTE: Because the plugin runs inside the host at the host's sample rate, Ampz brackets the chain with a high-quality resampler so the neural models always run at their native rate. **Any host sample rate works** — 44.1 kHz, 48 kHz, 96 kHz, and so on.

15.2 Loading Ampz in a Host

The plugin loads like any other AUv3 effect. The exact steps vary by host; the general flow is:

1. Open your host app (for example **AUM**, **Loopy Pro**, or **GarageBand for iOS**).
2. Add an **AUv3 effect** insert to a channel or track.
3. Choose **Ampz** from the **LNH Enterprises LLC** manufacturer group.
4. The Ampz interface opens inside the host — build your rig exactly as you would in the standalone app.

WARNING: Force-quit the standalone Ampz app before using the plugin. The standalone app claims the audio input, which can prevent the plugin (running inside the host) from receiving signal. Swipe the standalone app away from the app switcher first.

15.3 Sharing Models, IRs, and Presets

The app and the plugin share a common storage area, so your library is available in both:

1. NAM models and cabinet IRs you import in the **standalone app** appear automatically inside the **plugin**.

2. The plugin can browse and apply the app's saved **presets** — including auto-loading the app's current preset — so a rig you dialed in standalone is one tap away inside a host.
3. The plugin can also **save** back into the library: tap **Save**, then tap again to confirm the slot. The rig, its scenes and its stomps appear in the standalone app's preset browser.

TIP: Import your `.nam` models and IRs using the standalone app (its Files browser and Tone3000 integration make importing easy), then simply select them inside the plugin.

15.4 Host Automation

Ampz exposes a **named parameter tree** to the host that is rebuilt automatically whenever you change the chain. Each active slot appears as its own parameter group with the effect's real parameter names plus a **Bypass** switch, so you can record and automate them from your DAW. A global group adds **Scene** (0 to 7) and the Mix block's **Level A**, **Pan A**, **Level B**, **Pan B**, **Crossfade**, **Delay** and **Polarity B**, so a host can switch scenes and ride the lanes.

NOTE: Parameter addresses and identifiers are a stable compatibility contract — they do not change between sessions, so host automation you record against a rig keeps working after you reopen the project.

15.5 Saving Plugin State

When you save your host project (or a host/AUv3 preset), Ampz stores the **entire rig** — the effects chain, the signal path, all eight scenes, every slot's NAM model reference, and IR references — so the project recalls exactly as you left it. Ampz supports AUv3 **user presets**, so you can also save and recall rigs from the host's own preset menu.

Ampz User Manual

Version 1.9.0 — October 2026

Developed by LNH Enterprises LLC

NAM Technology powered by NeuralAudio

For support and updates, visit the Ampz website.

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